

Chapter 7

Theory of NMR in fluid-saturated porous media

Following the earlier work on NMR relaxation of fluid in a confined cell or pore space by Wayne and Cotts [223] and Robertson [187], the experimental findings by Callaghan et al. [50, 51, 52] on the diffraction-like behavior for the magnetization of fluid-saturated porous media, and numerous experimental investigations of the NMR properties of fluid-saturated rocks, there have been many theoretical discussions about the nuclear spin relaxation and diffusion in restricted geometries (e.g. Refs. [45, 95, 165, 43, 18]). In this chapter, we attempt to present a coherent treatment of the subject in a rigorous manner. Some of the mathematical derivations are deferred to the Appendices in order to allow for a clearer presentation in the main text.

7.1 Spin relaxation and diffusion

Spatial diffusion of the nuclei that produce an NMR spin echo can have a crucial effect on the properties of these echoes: When the spin polarizing magnetic field depends on position, the Larmor precession frequency of the nuclear spin of a single diffusing atom keeps changing with time in an incoherent fashion. This leads to an enhancement of the transverse decay rate in a Hahn's spin echo experiment, performed on a uniform or unconstrained fluid, from $1/T_2$ to

$$\frac{1}{T_2} + D_p(\gamma\nabla H)^2\frac{\tau^2}{3}. \quad (7.1)$$

Here D_p is the diffusion constant, γ is the gyromagnetic ratio of the diffusing spin, ∇H is the local gradient of the polarizing field H , and τ is the time between the initial $\pi/2$ pulse and the π spin inversion pulse, and also between the latter pulse and the spin echo [201]. This expression was derived and explained in Section 2.5.1.

When the fluid in which the precessing spin diffuses is confined either in a small finite volume, or in a connected network of pores of a porous medium, further changes in the transverse relaxation rate occur. The diffusion problem now has many eigenmodes. Each mode decays exponentially in time with its own characteristic decay rate, which is an eigenvalue of the time dependent diffusion problem.

These eigenvalues are also affected by the presence of sinks in the fluid or at its boundaries, where the polarization of a diffusing particle can be destroyed. A uniform volume distribution of such sinks provides a constant extra additive contribution to the intrinsic decay rate $1/T_2$, and its effects can therefore be taken into account by using an altered value for T_2 .

More subtle are the effects of a surface distribution of such sinks at the boundaries of the fluid, i.e., at the interface between the pore space and the solid rock matrix. This occurs when there are paramagnetic ions at that interface. The effects of such a “surface absorption” of polarization density are described by a coefficient of surface absorption¹ ρ , whose dimensions are length/time. Obviously, the importance of ρ will depend on the (linear) pore size a , since the absorption at the interface can only occur if the diffusing particle has reached it.

The competition between diffusion and surface absorption leads to the concept of fast versus slow diffusion: The diffusion is *fast* if $\rho a/D_p \ll 1$, and is *slow* if $\rho a/D_p \gg 1$. Clearly, in a uniform fluid, one is always in the slow diffusion limit, since $a \rightarrow \infty$. One might therefore think that, in this limit, the diffusion will be less sensitive to the structure of the pore space. However, we shall see in Section 7.1.1 below that this conclusion is premature, and that the slow diffusion limit in fact has the potential, under certain conditions, of providing detailed information about the pore structure by studying the effects of diffusion on NMR spin echoes.

In order to estimate the magnitude of $\rho a/D_p$ in typical porous rocks, we

¹This constant is referred to as “surface relaxivity” in earlier chapters. Here, we use a more general term “surface absorption” in order to broaden the scope of applicability of the solution for the diffusion problem, which is useful in many other areas such as catalysis, gas absorption, etc.

note that ρ in sandstones is usually in the range 1–10 $\mu\text{m}/\text{sec}$ (see Section 4.1.2, Table 4.1, and Ref. [32]), while the mean pore diameter a is in the range 0.0015–0.015 cm [118].² Using the value $D_p = 2.5 \times 10^{-5} \text{ cm}^2/\text{sec}$ for the self diffusion coefficient of pure water, we get

$$\frac{\rho a}{D_p} = 0.006 - 0.6.$$

This range is almost entirely within the fast diffusion regime. Nevertheless, we will consider the entire range of possible values of the dimensionless parameter $\rho a/D_p$, including both slow and fast diffusion regimes. This is done in order to show how measurements of spin echoes can be used to obtain useful information about the effects of pore structure on the macroscopic diffusion, and subsequently also on the permeability to flow of brine or hydrocarbon fluids through that structure.

7.1.1 Diffusion in restricted geometries with a uniform polarizing field

When there is both a volume distribution ν and a surface distribution ρ of spin absorbing sinks, the spin density $m(\mathbf{r}, t)$ satisfies the following equations

$$\frac{\partial m}{\partial t} = \nabla \cdot (D_p \nabla m) - \nu m \quad \text{in the pore space,} \quad (7.2)$$

$$0 = \mathbf{n} \cdot D_p \nabla m + \rho m \quad \text{at the pore/matrix interface,} \quad (7.3)$$

$$m(\mathbf{r}, t = 0) = \frac{M(0)}{V_p} \quad \text{at the initial time.} \quad (7.4)$$

Here $\mathbf{n}$ is the outward pointing unit vector normal to the pore/matrix interface (i.e., it points out of the pore space and into the solid rock), while $M(0)/V_p$ is the initial spin density, assumed to be uniform in the pore space of volume V_p . We note that D_p was not taken outside (i.e., to the left of) the $\nabla \cdot$ operator in the differential equation: That will allow us to use these equations to describe a situation where D_p is not a constant, but varies with position. This is sometimes a useful option, as explained in Appendix A.

In general, the spin density is actually a complex vector quantity, which represents a non-equilibrium spin polarization produced by a uniform, resonant rf field which is applied to the system for a short period of time before

²Note that the values quoted for ρ in this reference are too large, by more than an order of magnitude, as shown by the more recent measurements described in Chapter 4.

$t = 0$. E.g., after a π pulse of the rf field, $M(0)/V_p$ is a real vector opposite to the fixed polarizing field H_0 , while after a $\pi/2$ pulse, $M(0)/V_p$ represents a complex vector, perpendicular to H_0 . Because the direction and complex phase of $m(\mathbf{r}, t)$ are not altered by the diffusive motion, we ignore those features when solving the equations for $m(\mathbf{r}, t)$.

Note that the diffusion problem is set up with only two parameters, ν and ρ . Thus the solution is applicable for either T_1 or intrinsic T_2 , where ν or ρ would have a subscript 1 or 2, depending on whether m represents a polarization that is parallel or transverse to the fixed uniform spin polarizing magnetic field.

However, the function $m(\mathbf{r}, t)$ will be unable to describe the enhanced relaxation rate of transverse polarization due to field gradients, as in Eq. (7.1), because we did not include the field gradient contribution $-i\gamma(\mathbf{r} \cdot \nabla H)m$ to the rate of change of $m(\mathbf{r}, t)$ in Eq. (7.2), as we did in the Torrey equation (2.25) of Section 2.5.1. The field gradient term leads to position and time dependent changes in the complex phase of $m(\mathbf{r}, t)$. The Torrey equation was solved there for the case of a uniform fluid, where that term was found to enhance the transverse relaxation rate, as described by Eq. (7.1). That treatment of the field gradient is only possible in the case of an infinite uniform fluid.

In order to consider the dephasing and the refocusing of spins in a T_2 measurement in *restricted geometries*, we need to use the diffusion propagator which will be introduced in Section 7.1.2. For now, we will confine ourselves to a discussion of intrinsic T_2 due to volume or surface sinks only, and assume that the field is homogeneous and that the phases of all spins are identical.

Diffusion eigenstates

The eigenstates $m_n(\mathbf{r}, t)$ of this problem are states where the spatial density maintains a fixed shape and only its overall magnitude decays exponentially with time, i.e.

$$m_n(\mathbf{r}, t) = \psi_n(\mathbf{r})e^{-\lambda_n t}.$$

The eigenfunctions $\psi_n(\mathbf{r})$ and eigenvalues λ_n satisfy the equations

$$-\nabla \cdot (D_p \nabla \psi_n) + \nu \psi_n = \lambda_n \psi_n \quad \text{in the pore space,} \quad (7.5)$$

$$\mathbf{n} \cdot D_p \nabla \psi_n + \rho \psi_n = 0 \quad \text{at the pore/matrix interface.} \quad (7.6)$$

The time dependent spin density or polarization density $m(\mathbf{r}, t)$ can be

expanded in terms of the eigenstates as

$$m(\mathbf{r}, t) = \sum_{n=0}^{\infty} A_n \psi_n(\mathbf{r}) e^{-\lambda_n t}. \quad (7.7)$$

Since the eigenfunctions are mutually orthogonal, the expansion coefficients A_n can be found from the initial uniform density $m(\mathbf{r}, t=0) = M(0)/V_p$ by (all the integrals here are over the volume V_p of the pore space)

$$A_n = \frac{M(0)}{V_p} \frac{\int dV \psi_n^*(\mathbf{r})}{\int dV |\psi_n(\mathbf{r})|^2}.$$

The total time dependent polarization is then given by

$$M(t) = \int dV m(\mathbf{r}, t) = M(0) \sum_{n=0}^{\infty} f_n e^{-\lambda_n t}, \quad (7.8)$$

where the relative intensities or absolute probabilities f_n are given by

$$f_n = \frac{1}{V_p} \frac{|\int dV \psi_n(\mathbf{r})|^2}{\int dV |\psi_n(\mathbf{r})|^2}, \quad (7.9)$$

and sum up to 1

$$\sum_{n=0}^{\infty} f_n = 1. \quad (7.10)$$

Infinite uniform fluid

In a uniform fluid that fills up all space, the lowest eigenvalue λ_0 , corresponding to the longest relaxation time $\tau_0 = 1/\lambda_0$, is equal to ν , since $\psi_0(\mathbf{r}) \equiv \text{const.}$ Therefore ν can always be taken as the bulk transverse relaxation rate $1/T_{2B}$ plus any additional relaxation rate ν_0 that is produced by a volume distribution of sinks. In the following discussion, we shall simply use ν to include all these bulk effects, i.e., $\nu = 1/T_{2B} + \nu_0$, because they do not play any role in the special relaxation properties of porous media. Each of the other eigenstates can be completely characterized by a wave vector $\mathbf{q}$, which can take any value. The eigenfunctions are $e^{i\mathbf{q}\cdot\mathbf{r}}$, and the eigenvalues are

$$\lambda_{\mathbf{q}} = \nu + D_p \mathbf{q}^2, \quad (7.11)$$

which is obtained by substituting $e^{i\mathbf{q}\cdot\mathbf{r}}$ for ψ_n in Eq. (7.5). Here, and in every other case where ν has a constant value, we can always ignore ν when calculating the eigenvalues λ_n , and add it to those eigenvalues at the end of the calculation.

In the case of a uniform fluid, the continuous spectrum of relaxation rates of Eq. (7.11) makes no contribution to the relaxation rate of the total polarization $M(t)$: The only nonzero expansion coefficients in Eqs. (7.7) and (7.8) are A_0, f_0 , since all the other eigenfunctions are orthogonal to ψ_0 , which is constant, hence their integrals vanish. Therefore

$$\frac{M(t)}{M(0)} = e^{-\lambda_0 t} = e^{-\nu t}. \quad (7.12)$$

Restricted geometries

In restricted geometries, this is no longer true. These eigenfunctions, ψ_n , reflect the characteristics of the confined geometries. However, depending on whether the system is in the fast or slow diffusion regime, the information about the microstructure of the pore space may not be extractable from measurements of $M(t)$.

Fast diffusion limit In the *fast diffusion limit* $\rho a/D_p \ll 1$, the boundary condition at the pore/matrix interface reduces to $\mathbf{n} \cdot \nabla m \cong 0$. Consequently $\psi_0(\mathbf{r}) \cong \text{const}$ and $f_0 \cong 1$ [from Eq. (7.9)], so that the relaxation of $M(t)$ is again a one exponential decay characterized by the slowest decay rate λ_0 , which can be found, approximately, by the following calculation:

$$\begin{aligned} (\lambda_0 - \nu) \int dV \psi_0 &= - \int dV \nabla \cdot (D_p \nabla \psi_0) = - \oint dS D_p \frac{\partial \psi_0}{\partial n} = \oint dS \rho \psi_0 \\ \Rightarrow \lambda_0 &\cong \nu + \frac{\oint dS \rho}{V_p} \equiv \nu + \bar{\rho} \frac{S_p}{V_p}, \end{aligned} \quad (7.13)$$

where we used Eqs. (7.5) and (7.6). The volume integrals here are over the pore space of volume V_p , while the surface integrals $\oint dS$ are over the pore/matrix interface of area S_p ; $\bar{\rho}$ is the average of ρ over the interface, and S_p/V_p is the surface-to-volume ratio of the pore space, which is a good quantitative measure for the inverse of the typical linear size a of that space: It is valid for isolated pores of finite volume, as well as for a connected pore space that extends to unlimited distances.

However, S_p/V_p contains *no information* at all about the *shape* of the pore space, in particular, it does not provide any information about the existence of bottlenecks in that space, such as narrow necks or narrow cracks which separate between larger pores and can drastically impede both diffusion and fluid flow. In the case of isolated pores with simple shapes, it is easy to write an explicit form for this ratio, e.g., for a sphere of radius a we have $S_p/V_p = 3/a$, while for a circular cylinder of radius a we have $S_p/V_p = 2/a$, and for a parallel-slab-shaped pore of thickness $2a$ we have $S_p/V_p = 1/a$.

Because $\psi_0(\mathbf{r})$ is almost independent of $\mathbf{r}$ inside the pore space, the probability f_0 is very close to 1. Therefore all other eigenstates, with decay rates greater than λ_0 , make a negligible contribution to $M(t)$. Consequently $M(t)$ exhibits simple exponential decay at a rate given by Eq. (7.13).

Slow diffusion limit For isolated pores with simple shapes (i.e., sphere, infinitely long circular cylinder, flat region between two parallel planes), the eigenfunctions and eigenvalues can be calculated quite explicitly, in terms of trigonometric or Bessel functions and their zeros [45]. From these solutions we learn that, in the *slow diffusion limit* $\rho a/D_p \gg 1$ and for fixed pore shape, all eigenvalues satisfy $\lambda_n = g_n^2 D_p/a^2$. The eigenfunctions are *universal functions* (i.e., their form depends only on the pore shape) of $g_n \mathbf{r}/a$, where the dimensionless constants g_n are of order 1 and depend on the eigenvalue index n . They also depend on the pore shapes in a non-trivial fashion. Therefore, even if D_p is known, a measurement of just one relaxation time is unable to provide detailed information about the size and shape of the pore space. A fortiori, interesting physical properties, like fluid permeability, will not be determined by such a measurement. Only a measurement of the *distribution of relaxation rates* of $M(t)$, i.e., of both λ_n and f_n for many different values of n , has a chance of providing information about the topology, and consequently about the fluid permeability, of the porous system. In the slow diffusion regime, many decay modes contribute significantly to the decay of $M(t)$. Hence a non-trivial distribution of relaxation rates can be obtained even by measurements of limited accuracy.

It should be stressed that, although the slow diffusion limit has a greater potential for providing detailed microstructural information, it is not desirable to have D_p/a^2 too small: If that quantity is much smaller than the relaxation rate due to volume effects, like the pure fluid relaxation rate $1/T_{2B}$ and the volume distribution of sinks ν_0 , then it will be difficult to resolve the

spectrum of microstructure dependent relaxation rates λ_n . Ideally, we would like to have

$$\frac{1}{T_{2B}} + \nu_0 < \frac{D_p}{a^2} \text{ and } \frac{\rho a}{D_p} > 1,$$

which entails

$$\frac{\rho}{a} > \frac{1}{T_{2B}} + \nu_0. \quad (7.14)$$

Using the ranges quoted above for ρ and a (see Section 7.1), we find

$$\frac{\rho}{a} = 0.007 - 0.7 \text{ sec}^{-1}$$

for the typical range of values of their ratio. Since $T_{2B} < T_{1B} \cong 1 \text{ sec}$ for the relaxation time of protons in pure water, this means that the inequality of Eq. (7.14) is not satisfied. Therefore measurements of the distribution of relaxation rates in a uniform polarizing field will not be useful as a means to obtain detailed information about microstructure of the pore space.

Initial rate of decay Another interesting quantity that can sometimes be measured (see, e.g., Section 3.2.2) is the initial slope of the decaying total polarization $\dot{M}(0)$, also denoted by $-M(0)/T_{in}$, where $1/T_{in}$ is the initial relaxation rate. Using Eqs. (7.2) and (7.3), we can write the following expression for the instantaneous decay rate of $M(t)$:

$$\begin{aligned} -\dot{M}(t) &= -\int dV \dot{m}(\mathbf{r}, t) = -\int dV [\nabla \cdot (D_p \nabla m) - \nu m] \\ &= -\oint dS D_p \mathbf{n} \cdot \nabla m + \int dV \nu m = \oint dS \rho m + \int dV \nu m. \end{aligned}$$

At the initial time $t = 0$, we have $m(\mathbf{r}, t = 0) = M(0)/V_p$ in the pore space, therefore

$$\frac{1}{T_{in}} \equiv -\frac{\dot{M}(0)}{M(0)} = \nu + \bar{\rho} \frac{S_p}{V_p}. \quad (7.15)$$

This is the same as λ_0 in the fast diffusion limit—see Eq. (7.13). In that limit, the initial decay rate of $M(t)$ persists for all times, whereas for the slow diffusion limit, it is only true for the initial decay.

Summary In the slow diffusion limit, and also in the intermediate regime between the fast and the slow diffusion regimes, there is a non-trivial distribution of relaxation times for $M(t)$. These relaxation times depend on the pore sizes and shapes in a way that also depends on the surface absorption coefficient ρ . Therefore, if both D_p and ρ are known, measurements of this distribution in these regimes can provide detailed information about the topology of the pore space, and consequently about the fluid flow permeability. By contrast, in the fast diffusion limit the decay of $M(t)$ is a simple exponential function at all times, with a fixed decay rate equal to the initial rate of decay. That rate contains no information about the shape of the pore space, other than the ratio S_p/V_p .

7.1.2 The diffusion propagator

Before considering the effect of diffusion of spins in a fluid-saturated porous medium where the magnetic field is non-uniform, it is useful to introduce the concept of “diffusion propagator” $G(\mathbf{r}, \mathbf{r}'|t) = G(\mathbf{r}', \mathbf{r}|t)$. This is the conditional probability density to find a particle at $\mathbf{r}'$ at time $t > 0$, if a particle was present at $\mathbf{r}$ at time $t = 0$. In fact, this is just the Green’s function for the diffusion problem in the pore space, and it satisfies the following equations (here we assume that D_p is constant throughout the pore space)

$$\frac{\partial G}{\partial t} = D_p \nabla^2 G - \nu G \quad \text{in the pore space,} \quad (7.16)$$

$$0 = \frac{\partial G}{\partial n} + \frac{\rho}{D_p} G \quad \text{at the pore/matrix interface,} \quad (7.17)$$

$$G(\mathbf{r}, \mathbf{r}'|t=0) = \delta^3(\mathbf{r} - \mathbf{r}'). \quad (7.18)$$

In Section 7.2 below it is shown that the average of this propagator can be found by measuring an NMR spin echo in a “pulsed-field-gradient-spin-echo (PFGSE)” experiment. This is a powerful experimental technique which we discussed in Chapter 2, and it is frequently used for measuring the microstructure and diffusion properties of porous media: Since the time and spatial dependence of G reflect the detailed microstructure of the pore space, such measurements can provide rather detailed information about that microstructure. Even the more conventional measurements of NMR spin echos in the presence of a fixed gradient of the polarizing field (i.e. “fixed-gradient-

spin-echo" or FGSE), provide some information about the microstructure and the diffusion in such media.

The PFGSE amplitude $M(\mathbf{k}, t)$, which is measured in a PFGSE experiment, is related to the diffusion propagator by [see Eq. (7.61) below]

$$M(\mathbf{k}, t) \equiv \frac{1}{V_p} \int_{V_p} dV \int_{V_p} dV' G(\mathbf{r}, \mathbf{r}'|t) e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')}. \quad (7.19)$$

Here t is the time separation between the field gradient pulses, V_p is the total volume of the pore space, and the wave vector $\mathbf{k}$ is simply related to the polarizing field gradient ∇H , the length of time δ during which the two field gradient pulses are applied, and the gyromagnetic ratio γ of the polarized spin

$$\mathbf{k} = \gamma \delta \nabla H. \quad (7.20)$$

The PFGSE experimental method was described in Section 2.5.2. More details about the specific experimental procedure for measuring $M(\mathbf{k}, t)$ and the significance of the various parameters are provided in Section 7.2 below. The microstructure of the pore space is assumed to be "macroscopically homogeneous and isotropic", therefore the average of $G(\mathbf{r}, \mathbf{r}'|t)$ over different realizations of that microstructure (i.e., the ensemble average) will depend only on $|\mathbf{r} - \mathbf{r}'|$. In that case, $M(\mathbf{k}, t)$ includes all the information needed to reproduce that average G . Also, the single volume integral

$$\int_{V_p} dV' G(\mathbf{r}, \mathbf{r}'|t),$$

as well as $M(\mathbf{k}, t)$, are dimensionless and of order 1.

Upon close examination of Eq. (7.19), it is not too difficult to realize that the PFGSE amplitude is simply the average, over all spins, of the phase factor acquired when the Larmor frequency was momentarily changed by turning on a uniform field gradient ∇H for two short periods of time, of length δ each and separated by a time interval of length t , before and after application of the spin flipping π pulse of rf radiation (also, see the discussion in Section 7.2 below). This is different from the regular Hahn spin echo experiment, in which the spins are always under the influence of a time independent field gradient, i.e., a "fixed-gradient-spin-echo (FGSE)". However, we shall see later that many important quantities of the porous media, including the FGSE, can be derived using the diffusion propagator and the PFGSE

amplitude. Thus, both of them are very useful in the theoretical treatment of spin diffusion in porous media.

The magnetization or spin density $m(\mathbf{r}, t)$ inside the pore space, given that $m(\mathbf{r}, 0) = M(0)/V_p$, can be found from

$$m(\mathbf{r}, t) = \frac{M(0)}{V_p} \int_{V_p} dV' G(\mathbf{r}, \mathbf{r}'|t). \quad (7.21)$$

The spatial integral of this quantity is just the total polarization $M(t)$, therefore

$$\frac{M(t)}{M(0)} = \frac{1}{V_p} \int_{V_p} dV \int_{V_p} dV' G(\mathbf{r}, \mathbf{r}'|t) = M(\mathbf{k} = 0, t). \quad (7.22)$$

This is just the special case of $M(\mathbf{k}, t)$ when $\mathbf{k} = 0$, i.e., for the uniform field case when $\nabla H = 0$.

If there is no absorption or decay of spin polarization, i.e. $\rho = \nu = 0$, then, using Eqs. (7.16) and (7.17), we get:

$$\begin{aligned} \frac{d}{dt} \int_{V_p} dV G(\mathbf{r}, \mathbf{r}'|t) &= \int_{V_p} dV D_p \nabla^2 G = \oint_{\partial V_p} dS D_p \frac{\partial G}{\partial n} = 0 \\ &\Rightarrow \dot{M}(t) \equiv 0. \end{aligned}$$

In this case the total probability for finding the diffusing spin somewhere in the pore space is constant in time and equal to 1, and $M(t) \equiv M(0)$.

By contrast, when ρ and/or ν are nonzero, that probability decreases monotonically to 0 with increasing t , and the total spin polarization also diminishes gradually as a result of the surface and volume sinks. A nonzero constant value of ν merely contributes the exponential factor $e^{-\nu t}$ to G , and this factor is independent of microstructure. Therefore it can be left out of all microstructure dependent calculations and only be brought in at the end as a multiplicative factor. By contrast, a nonzero value of ρ , even if it is constant, will contribute to both the time dependence and the spatial form of G .

Infinite uniform fluid

In a uniform pure fluid, the diffusion propagator has the well known Gaussian form

$$G_0(\mathbf{r}, \mathbf{r}'|t) = \frac{1}{(4\pi D_p t)^{3/2}} \exp \left[-\frac{(\mathbf{r} - \mathbf{r}')^2}{4D_p t} - \nu t \right]. \quad (7.23)$$

This can easily be verified by substitution in the diffusion equation (7.2) or (7.16). The subscript 0 is used to identify quantities related to the infinite uniform fluid.

Recalling the probabilistic significance of G_0 , it is easy to calculate the mean square distance travelled by a particle (or “mean square displacement”), diffusing in a uniform fluid, during time t

$$\langle [\mathbf{r}(t) - \mathbf{r}(0)]^2 \rangle = \frac{\int dV' (\mathbf{r} - \mathbf{r}')^2 G_0(\mathbf{r}, \mathbf{r}'|t)}{\int dV' G_0(\mathbf{r}, \mathbf{r}'|t)} = 6D_p t. \quad (7.24)$$

Note that this result is independent of the existence of any uniform intrinsic relaxation rate $1/T_2$ or uniform volume distribution of sinks ν . That is so because particles that get absorbed during the time t do not contribute to the mean square displacement. In such a system, the PFGSE amplitude can be obtained by substituting Eq. (7.23) into Eq. (7.19):

$$M_0(\mathbf{k}, t) = e^{-D_p \mathbf{k}^2 t - \nu t}. \quad (7.25)$$

Setting $\mathbf{k} = 0$ leads to

$$\frac{M_0(t)}{M(0)} = e^{-\nu t},$$

which is the same as Eq. (7.12).

Restricted geometries

For restricted geometries, the diffusion propagator no longer has a simple Gaussian form as in Eq. (7.23). It can always be expanded using the diffusion eigenstates:

$$G(\mathbf{r}, \mathbf{r}'|t) = \frac{1}{V} \sum_{\lambda} e^{-\lambda t} \psi_{\lambda}(\mathbf{r}) \psi_{\lambda}^*(\mathbf{r}'), \quad (7.26)$$

where V is the total volume of the system, including both pore space and rock matrix. The eigenvalues λ and eigenfunctions $\psi_{\lambda}(\mathbf{r})$ are obtained by solving the following eigenvalue problem

$$\lambda \psi_{\lambda} + D_p \nabla^2 \psi_{\lambda} = 0 \quad \text{inside the pore space,} \quad (7.27)$$

$$D_p \frac{\partial \psi_{\lambda}}{\partial n} + \rho \psi_{\lambda} = 0 \quad \text{at the pore/matrix interface.} \quad (7.28)$$

These equations are the same as Eqs. (7.5) and (7.6), but the eigenfunctions are now normalized, in addition to being mutually orthogonal,

$$\frac{1}{V} \int_{V_p} dV \psi_\lambda^* \psi_\nu = \delta_{\lambda\nu}. \quad (7.29)$$

From Eqs. (7.22) and (7.26), the total spin polarization $M(t)$ can be written as a sum of exponentially decaying terms

$$\frac{M(t)}{M(0)} = \frac{1}{\phi} \sum_\lambda e^{-\lambda t} \left| \frac{1}{V} \int_{V_p} dV \psi_\lambda(\mathbf{r}) \right|^2, \quad \phi \equiv \frac{V_p}{V}. \quad (7.30)$$

For a general disordered porous medium, there is no simple way to obtain these eigenstates and hence the diffusion propagator. However, for periodic porous media, it is possible to numerically compute both quantities for the case where $\rho = 0$, as well as for the case where $\rho \neq 0$. This is illustrated in Appendix A, where we describe some calculations of the eigenstates, and of $M(\mathbf{k}, t)$ of Eq. (7.19), i.e., the spatial Fourier transform of the average G , for such media. Using these numerical solutions, many interesting quantities for porous media, such as PFGSE amplitude, enhanced T_2 relaxation in heterogeneous fields, etc., can be calculated, as we shall discuss in subsequent sections.

Although it is not possible to compute the diffusion propagator for a general disordered porous medium, it is possible to obtain its asymptotic behavior in certain limiting cases. From those asymptotic results the polarization decay $M(t)$ can also be calculated.

Short time behavior For short times t , i.e., $D_p t \ll a^2$ and $\rho t \ll a$, where a is a characteristic linear size of the pore space, the changes in the form of $G(\mathbf{r}, \mathbf{r}'|t)$ can be calculated to leading order in the small quantities $D_p t/a^2$ and $\rho t/a$ —see Appendix B. For $M(t)/M(0)$ this leads to the following short time asymptotics (we omit the factor $e^{-\nu t}$ here; see Eq. (B.3) in Appendix B and Ref. [167])

$$\frac{M(t)}{M(0)} \cong \begin{cases} 1 - \frac{S_p}{V_p} \rho t & \text{for small } \rho \ll (D_p/t)^{1/2}, \\ 1 - 2 \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} & \text{for large } \rho \gg (D_p/t)^{1/2}. \end{cases} \quad (7.31)$$

The first line here, which is always valid for sufficiently small ρ , agrees with results obtained earlier for $1/T_{in}$ —see Eq. (7.15). The second line shows

that even when ρ is very large, the short time decay of $M(t)$ depends only on the ratio S_p/V_p , and is insensitive to the shape of the pore space. In particular, it does not distinguish between a connected pore space and a collection of isolated voids, if they happen to have the same S_p/V_p ratio.

Long time behavior: The macroscopic stationary diffusion coefficient For sufficiently long times, and for $\mathbf{k}$ that is not too large, the PFGSE amplitude has a form similar to that of the uniform fluid function $M_0(\mathbf{k}, t)$ of Eq. (7.25), but with the free fluid diffusion constant D_p replaced by the “bulk effective (or macroscopic) stationary diffusion coefficient” D_e

$$\frac{M(\mathbf{k}, t)}{M(\mathbf{k} = 0, t)} \cong e^{-D_e \mathbf{k}^2 t} \text{ for } D_p t \gg a^2 \text{ and } |\mathbf{k}a| \ll 1. \quad (7.32)$$

Low lying eigenvalues and electrical conductivity The value of D_e is determined by the low lying eigenstates of the diffusion problem. This macroscopic diffusion coefficient is related to the eigenvalue spectrum of the diffusion equation. However, in contrast with the case of the uniform fluid, the eigenfunctions are no longer simple plane waves, and the eigenvalues do not have the simple form of Eq. (7.11).

When $\rho = 0$, the lowest eigenfunction is still a constant, but when $\rho \neq 0$ it is a non-constant function $\psi_0(\mathbf{r})$. All we can say, in general, about $\psi_0(\mathbf{r})$ is that it is real and positive throughout the connected part of the pore space, and that it is *macroscopically uniform*, namely, it does not exhibit any tendency to be localized in some small part of the pore space [21]. The detailed form of $\psi_0(\mathbf{r})$ depends on the shape of the pore space, and the same is true for the other eigenfunctions. However, the *low lying eigenstates* can be obtained, approximately, in a somewhat general fashion once $\psi_0(\mathbf{r})$ is known. Those states are characterized by a wave vector $\mathbf{q}$, as in the case of the uniform fluid, and have the following form when $|\mathbf{q}|a \ll 1$ [21]:

$$\psi_{\mathbf{q}}(\mathbf{r}) = \psi_0(\mathbf{r}) e^{i \sum_{\alpha} q_{\alpha} \phi_{\alpha}(\mathbf{r})} + \mathcal{O}(\mathbf{q}^2), \quad (7.33)$$

$$\lambda_{\mathbf{q}} = \lambda_0 + D_e \mathbf{q}^2 + \mathcal{O}(\mathbf{q}^4). \quad (7.34)$$

Here $\phi_{\alpha}(\mathbf{r})$ is the electrical potential of a conductivity problem in the pore space, with local conductivity equal to $\psi_0^2(\mathbf{r})\sigma_p$ and an average electric field

equal to $\mathbf{e}_\alpha$ (the unit vector along the α axis), and σ_p is the conductivity of the pore fluid. This leads to

$$\frac{D_e}{D_p} = \frac{\int_{V_p} dV \psi_0^2 (\nabla \phi_\alpha)^2}{\int_{V_p} dV \psi_0^2}. \quad (7.35)$$

Here we are assuming, as always, that D_e has the same value in all directions, i.e., that the microstructure is macroscopically isotropic. This expression can easily be generalized to cases where the microstructure is anisotropic, and hence the bulk effective diffusion coefficient D_e becomes a second rank tensor instead of a scalar quantity [21].

The value of D_e when $\rho = 0$ is closely related to the value of the macroscopic or bulk effective electrical conductivity σ_e of the fluid-saturated porous medium:

$$\frac{D_e}{D_p} = \frac{1}{\phi} \frac{\sigma_e}{\sigma_p} = \frac{1}{\phi F}, \quad (7.36)$$

where $F \equiv \sigma_p/\sigma_e$ is the “formation factor”. Note that here $\phi \equiv V_p/V$ denotes the volume fraction of the pore space or the porosity, and *not* the electrical potential ϕ_α of Eq. (7.35).

An important property of D_e is that it is always less than D_p [21]. Again, a measurement of $M(\mathbf{k}, t)$ for long times and small $\mathbf{k}$ does not yield any detailed information about the pore space.

When ρ is negligible, $\psi_0(\mathbf{r})$ is constant in the pore space. In that case, σ_e is the *actual electrical conductivity* of the fluid filled porous medium (assuming that the rock matrix does not conduct), which can be measured in standard ways. The measurement of D_e will provide us, however, with an independent method for obtaining the formation factor as a verification of the standard electrical measurement, if ϕ and D_p are known.

It is worth noting that, when $\rho = 0$, D_e/D_p is independent of the absolute size scale a of the pore space, i.e., if the entire microstructure is uniformly expanded or contracted by some constant factor, both D_e and σ_e remain unchanged. This is no longer true when $\rho \neq 0$. In the latter case, D_e does depend on the absolute size scale of the pore space. For this reason, measurements of D_e can then provide more information about the microstructure than when $\rho = 0$.

We note that the low lying eigenvalues of the *porous medium* form a continuous spectrum that looks much like the continuous spectrum of all the eigenvalues of the diffusion problem in a *uniform fluid* —compare Eqs. (7.11)

and (7.34). This is why measurement of the asymptotic long time behavior of the macroscopic diffusion can only provide us with the two quantities λ_0 and D_e , and why measurements at finite times are needed to obtain more detailed information about the microstructure.

Time dependent diffusion coefficient Because $G(\mathbf{r}, \mathbf{r}'|t) = G(\mathbf{r}', \mathbf{r}|t)$, therefore $M(-\mathbf{k}, t) = M(\mathbf{k}, t)$. Thus, for small $\mathbf{k}$, $M(\mathbf{k}, t)$ must be a quadratic function of $\mathbf{k}$. One can define a “time dependent diffusivity” $D(t)$, such that [167]:

$$M(\mathbf{k}, t) \cong M(0, t)e^{-D(t)\mathbf{k}^2 t} \text{ for } |\mathbf{k}|a \ll 1. \quad (7.37)$$

The function $D(t)$ starts out from $D(0) = D_p$, and ends up at $D(\infty) = D_e < D_p$, decreasing monotonically with increasing t . Figure 7.1 shows results of calculations of $D(t)$ for a pore space of a periodic simple cubic array of solid spheres. Neighboring spheres in this array can be either non-overlapping, partially overlapping, or just touching. In the case of just touching spheres, the porosity is $\phi = 1 - \pi/6 \cong 0.4764$. Results are shown for a variety of different porosities, ranging from non-overlapping to partially overlapping neighboring spheres, and for a variety of values of ρ .

As in the case of $M(t)$, short time asymptotics can be applied to get [see Eq. (B.4) in Appendix B]

$$\frac{D(t)}{D_p} \cong \begin{cases} 1 - \frac{4}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} & \text{for } \rho \ll (D_p/t)^{1/2}, \\ 1 - \frac{2}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} & \text{for } \rho \gg (D_p/t)^{1/2}. \end{cases} \quad (7.38)$$

Interestingly, to the order exhibited here, both results are independent of ρ .

Experiments have been conducted to measure the time dependent diffusion coefficient $D(t)$ as a function of time for water-saturated rocks [108, 139]. The signal usually diminishes before $D(t)$ reaches its asymptotic value $D_e = D_p/F\phi$. Later investigations [148, 149], using both laser-spin-polarized and thermally spin-polarized xenon gas, did manage to reach the asymptotic regime and verified this relation, as well as the short time expression in the first line of Eq. (7.38). Figure 7.2 shows results from those experiments.

While the short time and long time behavior of $D(t)$ can only give us values for S_p/V_p and D_e , its behavior at intermediate times is sensitive to other details of the pore structure. Thus, measurements of $D(t)/D_0$ at intermediate times have the potential of providing further information about those microstructural details. Section 7.1.3 provides an illustration of this idea.

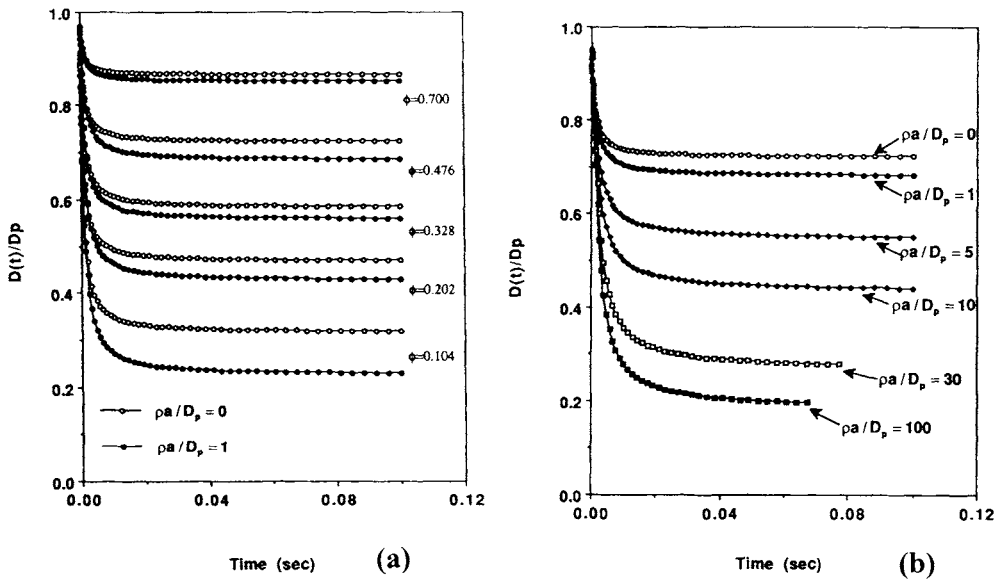


Figure 7.1: Bulk effective time dependent diffusion coefficient $D(t)/D_p$, plotted vs. time t , for a porous medium which is the spatial complement of a periodic, simple cubic array of solid spheres. (a) For various porosities ϕ and two values of the surface absorption coefficient: $\rho = 0$ and $\rho a/D_p = 1$. (b) For touching spheres ($\phi = 0.476$) and various values of $\rho a/D_p$ (after Fig. 3 of Ref. [18]).

7.1.3 Dephasing of the precessing spins

We now discuss the effects of diffusion on the dephasing of a precessing spin in the presence of a nonuniform magnetic field. Such a field can result from a uniform field gradient, intentionally applied in order to achieve some spatial resolution in the spin echo experiment, or it can result from unintentional and uncontrollable random spatial variations of the local magnetic susceptibility. The latter are mostly due to the inherent heterogeneity of the porous rock, i.e., the fact that the magnetic susceptibility χ has different values in the pore fluid and in the rock matrix.

The precession of a spin in a polarizing magnetic field directed along z is conveniently described by the time dependent expectation value $s_+(t)$ of the quantum mechanical spin operator $\hat{s}_x + i\hat{s}_y$. This quantity satisfies a

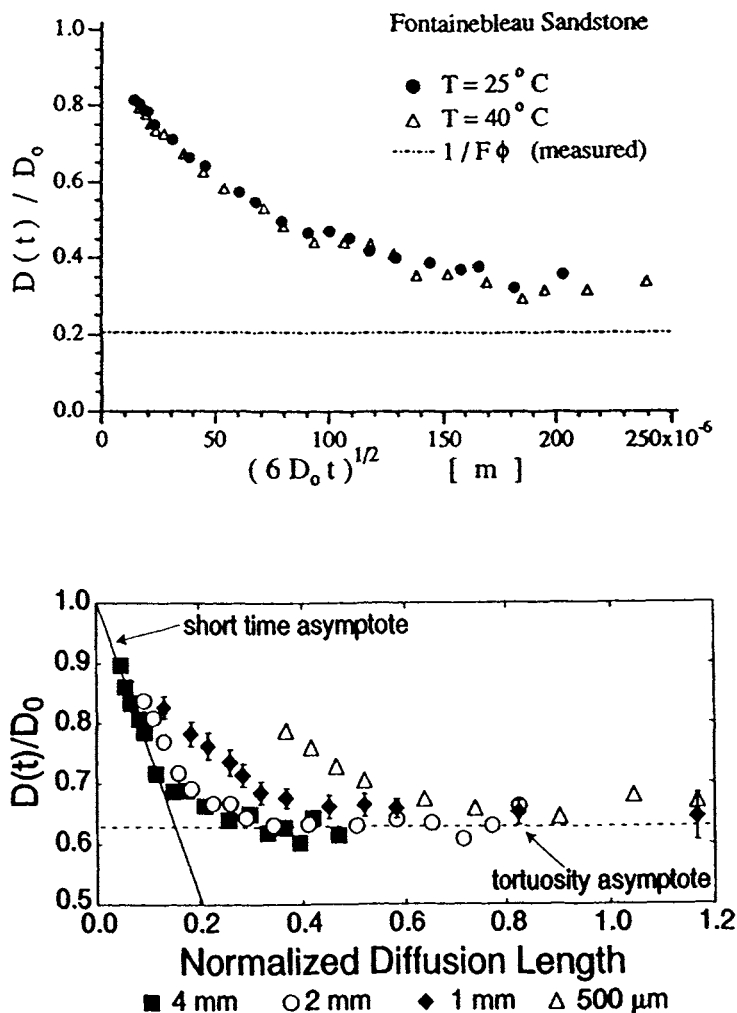


Figure 7.2: The figure on the top shows the bulk effective time dependent diffusion coefficient $D(t)/D_0$, plotted versus $(6D_0t)^{1/2}$, for a Fontainebleau sandstone, where the asymptotic line, $1/F\phi$, was measured independently using electrical method (From Hürlimann et al. [108]). The figure on the bottom shows the bulk effective time dependent diffusion coefficient measurements for thermally polarized xenon gas imbibed in randomly packed spherical glass beads of various sizes, where the asymptotic limit is reached (From Mair et al. [148, 149]).

Langevin type equation of motion

$$\frac{\partial s_+}{\partial t} = -i\gamma (H_0 + \delta H[\mathbf{r}(t)]) s_+ - \left(\frac{1}{T_{2B}} + \nu_0 \right) s_+ + F(t). \quad (7.39)$$

Here γ is the gyromagnetic ratio and we have written the polarizing field as the sum of a fixed uniform field H_0 and a position dependent variation $\delta H[\mathbf{r}(t)]$ —the latter contribution depends on the time dependent position $\mathbf{r}(t)$ of the diffusing spin; $1/T_{2B} + \nu_0$ is the transverse relaxation rate, which is a sum of the bulk relaxation rate $1/T_{2B}$ and a possible contribution ν_0 due to a constant volume distribution of sinks; $F(t)$ is a rapidly fluctuating random force term (i.e., noise). Without the latter term, which mimics the microscopic origins of the intrinsic relaxation rate, the solution of this equation is

$$s_+(t) = s_+(0)e^{-i\gamma H_0 t - (1/T_{2B} + \nu_0)t} \exp \left(-i\gamma \int_0^t dt \delta H[\mathbf{r}(t)] \right). \quad (7.40)$$

The first exponential factor in this expression includes the rapid precession of the polarized spin, at the Larmor angular frequency $\omega_c = \gamma H_0$ due to the main background polarizing field H_0 . It also includes the fixed background relaxation rate $1/T_{2B} + \nu_0$. The second exponential factor contains an extra phase $\Phi(t) = \gamma \int_0^t dt \delta H[\mathbf{r}(t)]$, accumulated by the precessing spin due to the position dependent variations in the polarizing field, which cause similar variations in the local Larmor frequency. The fact that this phase is different for different spins is responsible for an enhancement in the magnetization decay for the whole system.

Spin echo A spin echo experiment is best described in a reference frame that rotates around H_0 at the Larmor angular frequency $\omega_c = \gamma H_0$. In that frame, $s_+(t)$ is stripped of this rapid fixed rate of precession and becomes

$$s_{R+}(t) \equiv s_+(t)e^{i\omega_c t} = s_+(0)e^{-(1/T_{2B} + \nu_0)t} \exp \left(-i\gamma \int_0^t dt \delta H[\mathbf{r}(t)] \right). \quad (7.41)$$

In order to produce a spin echo at time 2τ , a short π pulse of a resonant rf field is applied at time τ after the initial $\pi/2$ pulse, in the xy plane. The π pulse changes $s_{R+}(t)$ into its complex conjugate. At time 2τ , the total extra phase shift, produced by δH , becomes

$$\Phi(2\tau) = -\gamma \int_0^\tau dt \delta H[\mathbf{r}(t)] + \gamma \int_\tau^{2\tau} dt \delta H[\mathbf{r}(t)]. \quad (7.42)$$

The sign difference between the phase accumulated before and after the π pulse is due to the complex conjugation of $s_{R+}(\tau)$ caused by that pulse. If the position of the spin $\mathbf{r}(t)$ does not change during the time interval $[0, 2\tau]$, then the two integrals exactly cancel each other and $\Phi(2\tau) = 0$: The dephasing is totally cancelled, and that revives the previously decayed rf radiation at frequency ω_c , and forms the spin echo. The amplitude of that echo is attenuated not only due to the uniform decay rate $1/T_{2B} + \nu_0$, but also due to the interface absorption [see Eq. (7.31)] and the diffusion [see Eq. (7.32)]. In this connection, recall that $M(\mathbf{k}, t)$ represents the amplitude of the spin echo in a PFGSE experiment.

In general, the time dependence of that decay is not well described by a simple exponential, except at very long times and small $\mathbf{k}$, or in the fast diffusion limit. Nevertheless, we sometimes refer to the instantaneous decay rate of the spin echo amplitude *in the absence of any magnetic field heterogeneity effects* as “intrinsic $1/T_2$ ”.

Effects of diffusion using the propagator Because of the diffusive motion, the two terms in Eq. (7.42) do not cancel each other when the field is heterogeneous. Moreover, the degree of non-cancellation differs for spins that move along different paths $\mathbf{r}(t)$. In order to calculate the effects due to this motion, we need to evaluate the average of $e^{-i\Phi(2\tau)}$ over all the spins in the system.

The presence of interface absorption affects the spin echo amplitude in two ways: (a) Even in the absence of any field heterogeneity, $M(t)$ is decreased when $\rho \neq 0$, therefore any spin echo will be similarly attenuated. Eq. (7.31) shows this effect in the limit of very short times. In general, this attenuation is not described by a simple exponential—see Eq. (7.30). (b) A nonzero value of ρ affects the diffusion propagator, and through it the average of $e^{-i\Phi(2\tau)}$. For example, $D(t)$ decreases with increasing ρ —see Fig. 7.1.

In the rotating reference frame, the field heterogeneity is always a small perturbation. Therefore the effect (a) above can be calculated by ignoring that heterogeneity, while the effect (b) can be calculated using the diffusion propagator, which is defined for the pore space *independently of any magnetic field*. The microstructure of the pore space, as well as the interface absorption, enter in the calculation of both effects.

Alternative approach An alternative approach to this calculation can start from Torrey's partial differential equation (2.25). That equation would then have to be solved with the boundary condition of Eq. (7.3) at the pore/matrix interface—a very difficult task. The approach described both above and below avoids that difficulty by treating the field heterogeneity as a *small perturbation*.

Cumulant expansion

One way to calculate the average $\langle e^{-i\Phi(2\tau)} \rangle$ is by expanding its natural (base e) logarithm in a series of cumulant averages $\langle \rangle_c$ of powers of $\Phi(2\tau)$, which will henceforth be abbreviated as Φ :

$$\ln \langle e^{-i\Phi} \rangle \equiv \langle e^{-i\Phi} - 1 \rangle_c = -i\langle \Phi \rangle_c - \frac{1}{2}\langle \Phi^2 \rangle_c + \frac{i}{6}\langle \Phi^3 \rangle_c + \frac{1}{24}\langle \Phi^4 \rangle_c + \dots$$

These cumulant averages can be expressed in terms of simple averages $\langle \rangle$ (see, e.g., Ref. [147]):

$$\begin{aligned} \langle \Phi \rangle_c &= \langle \Phi \rangle, \\ \langle \Phi^2 \rangle_c &= \langle \Phi^2 \rangle - \langle \Phi \rangle^2 \equiv \langle (\Delta\Phi)^2 \rangle, \quad \Delta\Phi \equiv \Phi - \langle \Phi \rangle, \\ \langle \Phi^3 \rangle_c &= \langle (\Delta\Phi)^3 \rangle, \\ \langle \Phi^4 \rangle_c &= \langle (\Delta\Phi)^4 \rangle - 3\langle (\Delta\Phi)^2 \rangle^2, \text{ etc.} \end{aligned}$$

When the distribution of values of Φ is Gaussian, then all cumulants beyond $\langle \Phi^2 \rangle_c$ vanish. For non-Gaussian distributions, the magnitudes of $\langle \Phi^3 \rangle_c$, $\langle \Phi^4 \rangle_c$, etc. are a measure of the deviation from Gaussian shape. The cumulant expansion is useful mainly if those deviations are small.

Averages over possible paths $\mathbf{r}(t)$ of a diffusing particle can be calculated by using the diffusion propagator $G(\mathbf{r}, \mathbf{r}'|t)$. Detailed calculations of such averages have been performed only on periodic microstructures and when $\rho = 0$. The odd order cumulants vanish when $\rho = 0$, including the first cumulant $\langle \Phi \rangle_c = \langle \Phi \rangle$, because time reversed paths have identical probabilities [19]. It follows that the distribution of Φ is approximately Gaussian, and is characterized by the variance of Φ , which is equal to its mean square $\langle \Phi^2 \rangle$ when $\langle \Phi \rangle = 0$, if we can show that all higher order cumulant averages are small compared to $\langle \Phi^2 \rangle$.

Periodic microstructures In order to illustrate how this is done, we note that in a periodic microstructure, the eigenstates $\psi_\lambda(\mathbf{r})$ have a Bloch-Floquet form, namely,

$$\psi_\lambda(\mathbf{r}) \rightarrow \phi_{n\mathbf{q}}(\mathbf{r})e^{i\mathbf{q}\cdot\mathbf{r}}, \quad (7.43)$$

$$\lambda \rightarrow \lambda_{n\mathbf{q}}, \quad (7.44)$$

where $\phi_{n\mathbf{q}}(\mathbf{r})$ has the same periodicity as the microstructure and $\mathbf{q}$ is a wave vector in the first Brillouin zone of the reciprocal space. The subscript n characterizes the different bands of eigenvalues $\lambda_{n\mathbf{q}}$, which are continuous functions of the $\mathbf{q}$ -vector. We have used here, for the first time, the notation $\phi_{n\mathbf{q}}$, $\lambda_{n\mathbf{q}}$ to distinguish the $\rho = 0$ eigenfunctions and eigenvalues from the more general quantities $\psi_{n\mathbf{q}}$, $\mu_{n\mathbf{q}}$ of a periodic porous medium with arbitrary, nonzero values of ρ .

The eigenvalues of the lowest band have the form

$$\lambda_{0\mathbf{q}} = \lambda_{00} + D_e \mathbf{q}^2 + \mathcal{O}(\mathbf{q}^4) \text{ for } |\mathbf{q}|a \ll 1. \quad (7.45)$$

The diffusion propagator now has the following form

$$G(\mathbf{r}, \mathbf{r}'|t) = \frac{1}{V} \sum_{n\mathbf{q}} e^{-\lambda_{n\mathbf{q}}|t|} \phi_{n\mathbf{q}}(\mathbf{r}) \phi_{n\mathbf{q}}^*(\mathbf{r}') e^{i\mathbf{q}\cdot(\mathbf{r}-\mathbf{r}')}, \quad (7.46)$$

where $\lambda_{n\mathbf{q}}$ are the eigenvalues of the n -th band and V is the total volume of the pore space and matrix.

To compute $\langle \Phi^2 \rangle$, we note that

$$\begin{aligned} \frac{1}{2\gamma^2} \langle \Phi^2 \rangle &= \left(2 \int_\tau^{2\tau} dt_2 \int_\tau^{t_2} dt_1 - \int_\tau^{2\tau} dt_2 \int_0^\tau dt_1 \right) \\ &\quad \cdot \langle \delta H[\mathbf{r}(t_2)] \delta H[\mathbf{r}(t_1)] \rangle. \end{aligned} \quad (7.47)$$

The average over diffusion paths is

$$\begin{aligned} &\langle \delta H[\mathbf{r}(t_2)] \delta H[\mathbf{r}(t_1)] \rangle \\ &= \frac{1}{V_p} \int_{V_p} dV_1 \int_{V_p} dV_2 G(\mathbf{r}_2, \mathbf{r}_1|t_2 - t_1) \delta H(\mathbf{r}_2) \delta H(\mathbf{r}_1). \end{aligned} \quad (7.48)$$

There are two sources for the field deviation δH :

$$\delta H(\mathbf{r}) = \mathbf{r} \cdot \nabla H + \delta H_{\text{susc}}(\mathbf{r}), \quad (7.49)$$

where the first term represents a uniform, time independent gradient of the spin aligning magnetic field, while the second term is the result of the fluid-matrix magnetic susceptibility difference $\Delta\chi \equiv \chi_p - \chi_m$ (χ_p , χ_m are the magnetic susceptibilities of pore fluid and rock matrix, respectively). The latter is the primary source of field heterogeneities at the pore scale.

Susceptibility difference contribution

The portion of δH that is due to the magnetic susceptibility contrast is given by [19]

$$\delta H_{\text{susc}}(\mathbf{r}) \cong 4\pi\Delta\chi H_0 \sum_{\mathbf{g} \neq 0} \cos^2(\mathbf{g}, \mathbf{e}_z) \theta_{\mathbf{g}} e^{i\mathbf{g} \cdot \mathbf{r}}. \quad (7.50)$$

Here $\theta_{\mathbf{g}} \equiv \frac{1}{V} \int dV \theta_p(\mathbf{r}) e^{-i\mathbf{g} \cdot \mathbf{r}}$ [see Eq. (A.8)] is the $\mathbf{g}$ -Fourier expansion coefficient of $\theta_p(\mathbf{r})$, and $\theta_p(\mathbf{r})$ is the characteristic function of the pore space, equal to 1 when $\mathbf{r}$ is inside that space and to 0 otherwise. The sum is over all the nonzero vectors $\mathbf{g}$ of the reciprocal lattice appropriate to the spatial periodicity of the pore space, and $(\mathbf{g}, \mathbf{e}_z)$ represents the angle between $\mathbf{g}$ and the direction of the spin polarizing field, i.e., the z axis.

When this is substituted in the evaluation of $\langle \Phi^2 \rangle$, we get the contribution due to the magnetic susceptibility contrast between pore fluid and solid matrix

$$\langle \Phi^2 \rangle_{\text{susc}} = \frac{2}{\phi} (4\pi\gamma\Delta\chi H_0)^2 \sum_{n>0} b_n(\tau) |a_n|^2. \quad (7.51)$$

Here $b_n(\tau)$, a_n are quantities that depend on the $\mathbf{q} = 0$ eigenstate of the band n : $b_0(\tau) \equiv 0$ —that is why the $n = 0$ term is absent in the sum—while $b_n(\tau)$ for $n \neq 0$ has a simple explicit form with useful limits for extreme values of τ [19]:

$$\begin{aligned} b_n(\tau) &= \frac{2\tau}{\lambda_{n0}} - \frac{1}{\lambda_{n0}^2} (1 - e^{-\lambda_{n0}\tau}) (3 - e^{-\lambda_{n0}\tau}) \\ &\cong \begin{cases} \frac{2}{3} \lambda_{n0} \tau^3 & \text{when } \lambda_{n0}\tau \ll 1, \\ \frac{2\tau}{\lambda_{n0}} & \text{when } \lambda_{n0}\tau \gg 1. \end{cases} \end{aligned} \quad (7.52)$$

The a_n 's depend on the $\mathbf{q} = 0$ eigenfunctions $\phi_{n0}(\mathbf{r})$ and are dimensionless:

$$a_n = \sum_{\mathbf{g} \neq 0} \cos^2(\mathbf{g}, \mathbf{e}_z) \theta_{\mathbf{g}}^* \tilde{\psi}_{n0}(\mathbf{g}), \quad (7.53)$$

where $\tilde{\psi}_{n0}(\mathbf{g})$ is the $\mathbf{g}$ -Fourier expansion coefficient of $\theta_p(\mathbf{r})\phi_{n0}(\mathbf{r})$ [see Eq. (A.7)].

We note that, if the pore space is macroscopically isotropic, a_0 has a value that is independent of the detailed shape of that space [this is a consequence of the exact result $\phi_{00}(\mathbf{r}) = 1/\sqrt{\phi}$ —see Eq. (A.13)]

$$a_0 = \frac{1}{\sqrt{\phi}} \sum_{\mathbf{g} \neq 0} \cos^2(\mathbf{g}, \mathbf{e}_z) |\theta_{\mathbf{g}}|^2 = \frac{1}{3} \sqrt{\phi} (1 - \phi).$$

The sum of all the other $|a_n|^2$ then satisfies the following inequality [19]:

$$\sum_{n>0} |a_n|^2 = \sum_n |a_n|^2 - a_0^2 \leq \frac{1}{3}\phi(1-\phi)\frac{2+\phi}{3}.$$

Note that the lowest eigenstate $n = 0$, $\mathbf{q} = 0$ does not contribute to the sum in Eq. (7.51).

Fixed gradient contribution

The contribution from an externally applied, fixed, uniform magnetic field gradient $\nabla H \parallel z$ is obtained by the following calculation

$$\begin{aligned} \frac{1}{\gamma^2} \langle \Phi^2 \rangle_{\text{grad}} = & \left(2 \int_{\tau}^{2\tau} dt_2 \int_{\tau}^{t_2} dt_1 - \int_{\tau}^{2\tau} dt_2 \int_0^{\tau} dt_1 \right) \\ & \left(-\frac{1}{V_p} \right) \int_{V_p} dV_1 \int_{V_p} dV_2 [(\mathbf{r}_2 - \mathbf{r}_1) \cdot \nabla H]^2 G(\mathbf{r}_2, \mathbf{r}_1 | t_2 - t_1). \end{aligned} \quad (7.54)$$

The spatial integrals are evaluated by taking derivatives of the Fourier transform of G

$$\begin{aligned} -\frac{1}{V_p} \int_{V_p} dV_1 \int_{V_p} dV_2 [(\mathbf{r}_2 - \mathbf{r}_1) \cdot \nabla H]^2 G(\mathbf{r}_2, \mathbf{r}_1 | t_2 - t_1) \\ = \left(\nabla H \cdot \frac{\partial}{\partial \mathbf{k}} \right)^2 M(\mathbf{k}, t_2 - t_1) \Big|_{\mathbf{k}=0}, \end{aligned} \quad (7.55)$$

where

$$M(\mathbf{k}, t) \equiv \frac{1}{V_p} \int_{V_p} dV \int_{V_p} dV' G(\mathbf{r}, \mathbf{r}' | t) e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} \quad (7.56)$$

is the PFGSE amplitude.

The resulting additive contribution to the mean square of Φ from the fixed magnetic field gradient is

$$\langle \Phi^2 \rangle_{\text{grad}} = (\gamma \nabla H)^2 \left(\frac{4}{3} D_e \tau^3 + \frac{1}{\phi} \sum_{n>0} b_n(\tau) d_n \right), \quad (7.57)$$

where d_n is given by [19]

$$d_n = \left(\gamma \nabla H \cdot \frac{\partial}{\partial \mathbf{q}} \right)^2 |\tilde{\psi}_{n\mathbf{q}}(\mathbf{0})|^2 \Big|_{\mathbf{q}=0}.$$

From Eq. (7.52) it follows that, for very long times, only the first term of Eq. (7.57) is important and we get a result like Eq. (7.1), but with D_p replaced by

D_e . For intermediate times, the second term makes a significant contribution to the decay rate of the spin echo amplitude. That contribution depends on the $\mathbf{q} = 0$ diffusion eigenstates, and hence on the microstructure and on the value of D_p , in a non-trivial fashion. At very short times, the second term in Eq. (7.57) combines with the first term to give back the result of Eq. (7.1).

Summary of the theory

The spin echo decay rates, which result from putting the results of Eqs. (7.51) and (7.57) in the exponent of $e^{-\frac{1}{2}\langle\Phi^2\rangle}$, appear on top of the decay rate of the total spin polarization $M(t)$, which is present even if the polarizing field is completely uniform.

Eqs. (7.51) and (7.57), together with the second line of Eq. (7.52), show that at long times (i.e., $\lambda_{n0}\tau \gg 1$), a spectrum of exponential decays of the spin echo amplitude is produced, which depends on the diffusion eigenmodes. It therefore reflects details of the microstructure of the pore space. The precise relation between this distribution of decay rates and the microstructure is a complex one, even in the relatively simple case of periodic microstructures: Both the eigenvalues and the eigenfunctions of the diffusion problem participate in producing that relation.

Attempts were made, at times, to calculate the effects of susceptibility difference by deriving an average value for the spatially fluctuating field gradient, and using that in Eq. (7.1). That approach totally ignores the effects of the restricted geometry and the interface absorption on the diffusion process. In practice, this ad-hoc approximation appears to grossly overestimate the decay rate of the spin echos [80]. A somewhat better ad-hoc approximation is to consider the pore space as a collection of independent small pores and compute the field gradient in each one. A value of $\langle\Phi^2\rangle$ is then calculated for each pore separately, using Eq. (7.1). The results for individual pores are then averaged over the distribution of pore sizes and shapes—see Sections 4.2.3 and 4.2.4. This leads to much better agreement with experiments, though it relies heavily on a Gaussian assumption for the distribution of Φ^2 [80].

Clearly, the approach described here is better founded and more systematic. In particular, it allows the validity of the Gaussian approximation to be monitored, and it even allows one to go beyond that approximation. Its main disadvantage is the limitation to periodic microstructures.

Enhanced T_2 relaxation rate in CPMG experiments

The magnetic field inhomogeneities affecting T_2 measurements for porous media come from two sources. One is the field gradient externally applied to the system. It occurs over a much larger scale than the pore dimensions, and in the treatment described here we assume that it is uniform throughout the system. Another source of field inhomogeneities is the magnetic susceptibility difference between the pore fluid and solid matrix, which makes the *local field* non-uniform even when the *applied field* is accurately uniform. The resultant spatial variations are at the pore size scale and can be strongly affected by the local pore geometry. Assuming the phase angles of spins have a Gaussian distribution (which is a good approximation if $\langle \Phi^4 \rangle_c \ll \langle \Phi^2 \rangle$), the magnetization of the system at time 2τ is given by

$$\frac{M(2\tau)}{M(0)} = \exp \left(-\frac{2\tau}{T_2} - \frac{1}{2} \langle \Phi^2 \rangle_{\text{susc}} - \frac{1}{2} \langle \Phi^2 \rangle_{\text{grad}} \right), \quad (7.58)$$

where $\langle \Phi^2 \rangle_{\text{susc}}$ and $\langle \Phi^2 \rangle_{\text{grad}}$ are given by (7.51) and (7.57), respectively.

In a CPMG experiment, instead of just one π pulse, an entire sequence of π pulses is applied, with a time interval 2τ between sequential pulses. A corresponding train of echoes is observed, with gradually decreasing magnitudes [201]. For a periodic microstructure, it can be shown that the decay of that echo train is described by [19, 43]

$$\begin{aligned} \ln \frac{M(2n\tau)}{M(0)} + \frac{2n\tau}{T_2} = & -(\gamma \nabla H)^2 \left[\frac{2n\tau^3 D_e}{3} + \frac{1}{2\phi} \sum_{m>0} c_m(\tau) \frac{\partial^2}{\partial q^2} |\tilde{\psi}_{m\mathbf{q}}(\mathbf{0})|^2 \right]_{\mathbf{q}=\mathbf{0}} \\ & - \frac{1}{\phi} (4\pi\gamma\Delta\chi H_0)^2 \sum_{m>0} c_m(\tau) |a_m|^2, \end{aligned} \quad (7.59)$$

where $c_m(\tau)$ is given by

$$c_m(\tau) = \frac{2n\tau}{\lambda_{m0}} \left(1 - \frac{\tanh(\lambda_{m0}\tau)}{\lambda_{m0}\tau} \right) + \frac{(1 - e^{-\lambda_{m0}\tau})}{\lambda_{m0}^2 (1 + e^{-2\lambda_{m0}\tau})^2} [1 + (-)^n e^{-2n\lambda_{m0}\tau}]. \quad (7.60)$$

The enhancement of the $1/T_2$ relaxation rate is obtained by differentiating Eq. (7.59) with respect to elapsed time t , where $t = 2n\tau$, keeping all the other factors of τ fixed.

Numerical examples

Detailed calculations of $\langle \Phi^2 \rangle$ have only been made for model periodic microstructures with a simple unit cell—see Appendix A and Ref. [19]. The inverse problem, starting from the distribution of decay rates and trying to reconstruct the microstructure, has not been tackled even for those simple models. Moreover, details of the microstructure are often not important in themselves, but only inasmuch as they enable us to predict other macroscopic properties of the medium, such as permeability to fluid flow. That is why most attempts to connect diffusion properties and permeability have been empirical.

These calculations show that, under conditions that usually prevail in brine saturated porous rocks, $\frac{1}{24}\langle \Phi^4 \rangle_c$ is less than $\frac{1}{2}\langle \Phi^2 \rangle_c$. These statements are based upon results of calculations on periodic pore structures, like the ones shown in Figs. 7.3 and 7.4: Fig. 7.3 shows results of computations of $\frac{1}{2}\langle \Phi^2 \rangle_c$ and $\frac{1}{24}\langle \Phi^4 \rangle_c$, due to susceptibility difference $\Delta\chi$ between water and rock, for two representative values of that difference and for different values of a and τ . However, it is not uncommon to find brine-saturated rocks that contain strong paramagnetic impurities resulting in a large magnetic susceptibility contrast, and leading to non-Gaussian situation.

Fig. 7.4 shows similar plots due to a fixed field gradient ∇H . In both cases, the dependence of $\langle \Phi^2 \rangle_c$ on $\Delta\chi$ or ∇H is *quadratic*, while the dependence of $\langle \Phi^4 \rangle_c$ is *quartic*. That is why, when ∇H is increased from 1 Gauss/cm to 20 Gauss/cm, the relative importance of $\frac{1}{24}\langle \Phi^4 \rangle_{c,grad}$ increases sufficiently so as to make it greater than $\frac{1}{2}\langle \Phi^2 \rangle_{c,grad}$ for times greater than about 30 msec—see Fig. 7.4.

7.2 PFGSE and its applications

The pulsed-field-gradient-spin-echo (PFGSE) NMR method was first proposed in 1963 [157] and demonstrated experimentally in 1965 [205]. Basically, it is a spin echo experiment in which a uniform polarizing field H_0 is modified by applying the same uniform field gradient ∇H for a number of short time intervals during both the decay period and the reconstruction period of the NMR signal. Those field gradient pulses are set so that they would have a null effect on any precessing spin that does not change its position during the entire train of field gradient pulses. Thus, by comparing the spin echo

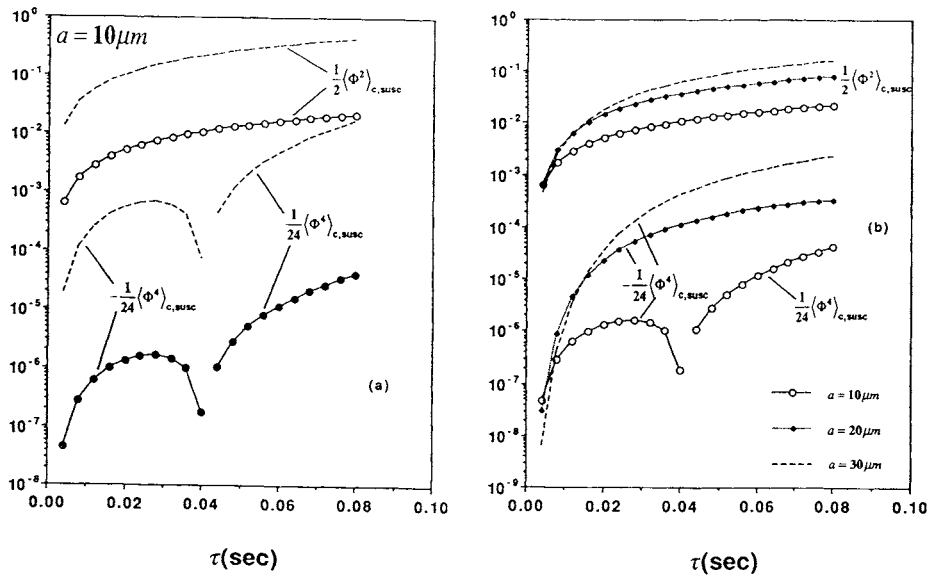


Figure 7.3: Results of $\frac{1}{2}\langle\Phi^2\rangle_{c,susc}$ and $\frac{1}{24}\langle\Phi^4\rangle_{c,susc}$ vs. τ for a brine filled periodic pore space made by embedding a simple cubic array of identical touching solid spheres of diameter a in the fluid. (a) $a = 10\mu m$, $|\Delta\chi| = 0.6 \times 10^{-6} \text{ emu/cm}^3$ (circles) and $|\Delta\chi| = 2.7 \times 10^{-6} \text{ emu/cm}^3$ (dashed lines). (b) $|\Delta\chi| = 0.6 \times 10^{-6} \text{ emu/cm}^3$ and three different values of a . Note that $\frac{1}{24}\langle\Phi^4\rangle_{c,susc}$ is positive for $a = 10\mu m$ when τ is greater than 40 ms, and negative for all other values of a and τ (after Fig. 4 of Ref. [19]).

with and without those field gradient pulses, information is obtained about motion of the precessing spins.

In the simplest type of PFGSE experiment, a single pulse of spatially uniform field gradient ∇H is first applied, for a short time interval δ between the initial rf $\pi/2$ pulse and the subsequent spin inverting rf π pulse. Then another identical ∇H pulse is applied between that π pulse and the subsequent spin echo. Fig. 2.9 shows a schematic representation of these events plotted vs. time (note that $\mathbf{g}$ is used to denote the field gradient ∇H in that figure).

The analysis of this experiment is based upon Eq. (7.42): The extra phase shift acquired by a single precessing spin is given by an integral over the deviation δH of the local polarizing field from its uniform background value $H_0 \parallel z$. That deviation now depends on time, because ∇H is turned

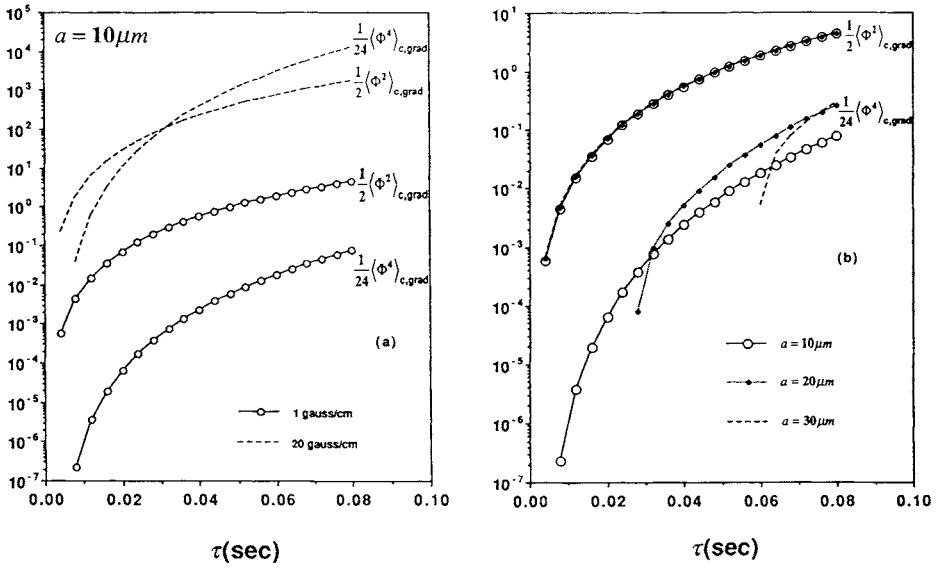


Figure 7.4: Results of $\frac{1}{2}\langle\Phi^2\rangle_{c,\text{grad}}$ and $\frac{1}{24}\langle\Phi^4\rangle_{c,\text{grad}}$ vs. τ for the same brine filled pore space as in Fig. 7.3, with an externally applied magnetic field gradient ∇H . (a) $a = 10 \mu\text{m}$ and two different values of magnetic field gradient. (b) one magnetic field gradient (1 Gauss/cm) for different values of a (after Fig. 5 of Ref. [19]).

on only for two short time intervals, of length δ each, but also because the spin changes its position $\mathbf{r}(t)$ with time. If the change in position during the time δ is negligible, then that phase shift is given by

$$\begin{aligned}\Phi &= -\gamma \int_0^\tau dt \delta H[\mathbf{r}(t), t] + \gamma \int_\tau^{2\tau} dt \delta H[\mathbf{r}(t), t] \\ &\cong -\gamma \delta (\delta H[\mathbf{r}(t_1), t_1] - \delta H[\mathbf{r}(t_2), t_2]) = \gamma \delta \nabla H \cdot [\mathbf{r}(t_2) - \mathbf{r}(t_1)] \\ &\equiv \mathbf{k} \cdot [\mathbf{r}(t_2) - \mathbf{r}(t_1)], \quad \mathbf{k} = \gamma \delta \nabla H,\end{aligned}$$

where t_1, t_2 are the moments at which the two gradient pulses are applied—in Fig. 2.9 the interval between those times is denoted by Δ . Note that the opposite signs of the phase shifts, caused by the two gradient pulses, is due to the fact that the quantum expectation value of $\hat{s}_x + i\hat{s}_y$, in the frame of reference which rotates around $H_0 \parallel z$ at the Larmor angular frequency γH_0 , is changed into its complex conjugate by the spin inverting π pulse which is applied between the two gradient pulses—see Fig. 2.9 and the discussion in

Section 7.1.3. Obviously, if $\mathbf{r}(t_2) = \mathbf{r}(t_1)$ then the phase shifts caused by the two field gradient pulses cancel each other. Any non-perfect cancellation is due to spatial motion of the spin between the times t_1 and t_2 .

The total effect on the spin echo signal from diffusive motion of all the precessing spins is found by averaging $e^{-i\Phi}$ over the distribution of diffusion paths $\mathbf{r}(t)$. This is done with the help of the diffusion propagator $G(\mathbf{r}, \mathbf{r}'|t)$, which is the probability density that a particle, which started out at $\mathbf{r}$, is found at $\mathbf{r}'$ after time t (that time is denoted by Δ in Fig. 2.9):

$$\langle e^{-i\Phi} \rangle = \frac{1}{V_p} \int_{V_p} dV \int_{V_p} dV' G(\mathbf{r}, \mathbf{r}'|t) e^{-i\mathbf{k} \cdot (\mathbf{r} - \mathbf{r}')} \Big|_{t=|t_2-t_1|} \equiv M(\mathbf{k}, |t_2 - t_1|). \quad (7.61)$$

More complicated sequences of field gradient pulses have also been used, at times, in order to achieve a better signal-to-noise ratio in measurements, and also for achieving other specialized goals [114].

Eq. (7.61) is just the time dependent spatial Fourier transform of the average diffusion propagator. $M(\mathbf{k}, t)$ is also known as the “PFGSE amplitude”, and can obviously be measured for values of $\mathbf{k}$ and t that are chosen by setting the experimental parameters. This capability has enormous potential for providing detailed information about the microstructure of the pore space, with a resolution that is determined by the range of $\mathbf{k}$ values that are accessed.

7.2.1 Diffraction-like behavior of PFGSE

Callaghan et al. [50] demonstrated very convincingly the diffraction-like behavior of PFGSE in an NMR experiment on a water filled porous medium, made by randomly and loosely packing a collection of monodispersed (i.e., same size) polystyrene spheres of diameter $16 \mu\text{m}$. In this experiment, a peak in $M(\mathbf{k}, t)$ was found to develop, as the time $t = t_2 - t_1$ increased up to 110 msec, at a value of $|\mathbf{k}|$ corresponding to a pore separation distance $2\pi/|\mathbf{k}|$ of about $16 \mu\text{m}$. In fact, this diffraction-like behavior can be shown theoretically to be a unique property of $M(\mathbf{k}, t)$. We shall briefly discuss some methods for calculating the PFGSE amplitude and some results of such calculations.

In Appendix A we show that, for a periodic pore space, the differential equation for the $\rho = 0$ diffusion eigenstates can be recast as a discrete matrix eigenvalue problem, where the eigenvectors are the Fourier expansion coefficients of the diffusion eigenfunctions—see Eq. (A.3). Since there are

infinitely many expansion coefficients, the matrix is discrete but infinite. By restricting the calculation to a finite subset of those expansion coefficients, the infinite matrix gets truncated to a finite size matrix, whose eigenvectors and eigenvalues can be found by standard numerical procedures. The eigenstates of the $\rho \neq 0$ diffusion problem are then expanded in the $\rho = 0$ eigenstates, and this leads to a discrete matrix eigenvalue problem for the $\rho \neq 0$ eigenstates. A truncated version of the latter problem is then solved using standard numerical procedures.

To compute the PFGSE amplitude, $M(\mathbf{k}, t)$, we recall that the diffusion propagator has the following general form—see Eq. (7.26):

$$G(\mathbf{r}, \mathbf{r}'|t) = \frac{1}{V} \sum_{\lambda} e^{-\lambda t} \psi_{\lambda}(\mathbf{r}) \psi_{\lambda}^*(\mathbf{r}').$$

When this is substituted in Eq. (7.61), we get

$$M(\mathbf{k}, t) = \frac{1}{\phi} \sum_n e^{-\lambda t} |\tilde{\psi}_{\lambda}(\mathbf{k})|^2, \quad (7.62)$$

where

$$\tilde{\psi}_{\lambda}(\mathbf{k}) = \frac{1}{V} \int_V dV \theta_p(\mathbf{r}) \psi_{\lambda}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{r}} = \frac{1}{V} \int_{V_p} dV \psi_{\lambda}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{r}}$$

is the Fourier transform of $\theta_p(\mathbf{r})\psi_{\lambda}(\mathbf{r})$, with $\psi_{\lambda}(\mathbf{r})$ normalized as in Eq. (7.29).

Periodic media

For periodic porous media, the diffusion propagator becomes [compare this with Eq. (7.46), which refers to the special case where $\rho = 0$]

$$G(\mathbf{r}, \mathbf{r}'|t) = \frac{1}{V} \sum_{n\mathbf{q}} e^{-\mu_{n\mathbf{q}}|t|} \psi_{n\mathbf{q}}(\mathbf{r}) \psi_{n\mathbf{q}}^*(\mathbf{r}') e^{i\mathbf{q} \cdot (\mathbf{r} - \mathbf{r}')},$$

where $\mu_{n\mathbf{q}}$ is an eigenvalue and $\psi_{n\mathbf{q}}(\mathbf{r})$ is the periodic part of the $\rho \neq 0$ Bloch-Floquet-type eigenfunction, normalized as in Eq. (7.29)—see Appendix A. $M(\mathbf{k}, t)$ is then given by the following expression

$$M(\mathbf{k}, t) = \frac{1}{\phi} \sum_n e^{-\mu_{n\mathbf{q}} t} |\tilde{\psi}_{n\mathbf{q}}(\mathbf{g}_{\mathbf{k}})|^2, \quad \mathbf{q} = \mathbf{k} - \mathbf{g}_{\mathbf{k}}, \quad (7.63)$$

where $\mathbf{g}_\mathbf{k} \equiv \mathbf{k} - \mathbf{q}$ is the reciprocal lattice vector that is closest to $\mathbf{k}$, and

$$\tilde{\psi}_{n\mathbf{q}}(\mathbf{g}) \equiv \frac{1}{V_a} \int_{V_p \cap V_a} dV \psi_{n\mathbf{q}}(\mathbf{r}) e^{-i\mathbf{g} \cdot \mathbf{r}}, \quad \sum_{\mathbf{g}} \tilde{\psi}_{n\mathbf{q}}^*(\mathbf{g}) \tilde{\psi}_{m\mathbf{q}}(\mathbf{g}) = \delta_{nm} \quad (7.64)$$

are the normalized eigenvectors of the diffusion problem. Also, V_a is the volume of a single unit cell of the periodic porous medium, including both pore space and rock matrix, while the integral is only over the volume of pore space in one unit cell. This expression for $M(\mathbf{k}, t)$ is valid irrespective of whether $\rho = 0$ or $\rho \neq 0$, even though the values of $\mu_{n\mathbf{q}}$ and $\tilde{\psi}_{n\mathbf{q}}(\mathbf{g})$, as well as the calculational procedure for obtaining these quantities, are quite different, as explained above. In the following, we will use the notation $\lambda_{n\mathbf{q}}$, $\phi_{n\mathbf{q}}(\mathbf{r})$ to denote the $\rho = 0$ limit of $\mu_{n\mathbf{q}}$, $\psi_{n\mathbf{q}}(\mathbf{r})$.

The extent to which $M(\mathbf{k}, t)$ includes detailed information about the microstructure is exhibited dramatically in the case of a periodic microstructure. Even though $G(\mathbf{r}, \mathbf{r}'|t)$ is not a periodic function, its Fourier transform $M(\mathbf{k}, t)$ exhibits peaks and valleys at characteristic values of $\mathbf{k}$ that are intimately connected to the spatial periodicity. Fig. 7.5 shows this behavior for $\mathbf{k}$ along the (100) direction in a microstructure which results when a simple cubic array of touching solid spheres is embedded in an otherwise uniform fluid, assuming $\rho = 0$. Fig. 7.6 shows similar results for the same system when $\rho a/D = 1$ and 10. The oscillatory behavior of $M(\mathbf{k}, t)$ is reminiscent of the periodic dependence of the eigenvalues $\mu_{n\mathbf{q}}$ upon $\mathbf{q}$ [see Figs. A.1 and A.2 in Appendix A], which is due to the periodic microstructure and follows from Bloch's theorem for such structures.

Properties of $M(\mathbf{k}, t)$

There are several things worth noting in connection with $M(\mathbf{k}, t)$ and Fig. 7.5: (a) The PFGSE amplitude is real, and it is even under space inversion, namely $M(-\mathbf{k}, t) = M(\mathbf{k}, t)$. This follows from the fact that G is real and symmetric $G(\mathbf{r}, \mathbf{r}'|t) = G(\mathbf{r}', \mathbf{r}|t)$. (b) From the inversion symmetry of the microgeometry it follows that

$$\theta_{\mathbf{g}} = \theta_{-\mathbf{g}} = \text{real} \quad (7.65)$$

$$\lambda_{n-\mathbf{q}} = \lambda_{n\mathbf{q}} \quad (7.66)$$

$$\tilde{\psi}_{n-\mathbf{q}}(\mathbf{g}) = \pm \tilde{\psi}_{n\mathbf{q}}(-\mathbf{g}) = \text{real}. \quad (7.67)$$

As a result, we only need to compute eigensolutions for $|\mathbf{q}|a$ from 0 up to the first Brillouin zone boundary π for any direction of $\mathbf{q}$, and retain all the

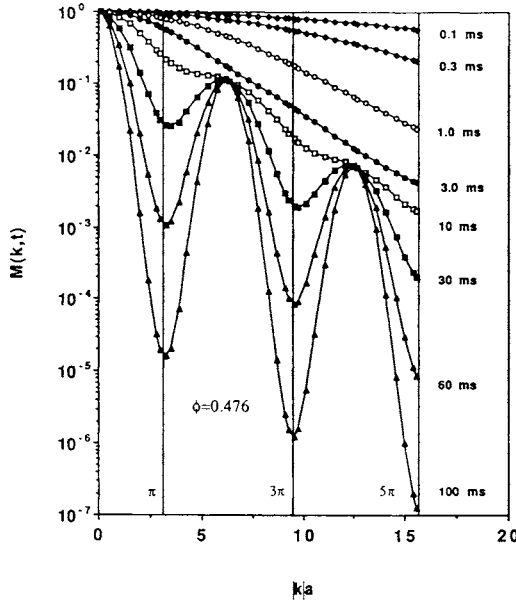


Figure 7.5: The PFGSE amplitude $M(\mathbf{k}, t)$ as function of $|\mathbf{k}|a$ along the (100) direction at various times for a simple cubic array of identical touching spheres with $\phi=0.476$ and $\rho = 0$. The Bragg planes at π , 3π , and 5π are also indicated (after Fig. 7 of Ref. [77]).

necessary weights $|\tilde{\psi}_{n\mathbf{q}}(\mathbf{g}_{\mathbf{k}})|^2$ in addition to $|\tilde{\psi}_{n\mathbf{q}}(\mathbf{0})|^2$. This allows us to plot $M(\mathbf{k}, t)$ beyond the first Brillouin zone boundary, in fact for any $\mathbf{k} = \mathbf{g}_{\mathbf{k}} + \mathbf{q}$. (c) For sufficiently long times t , $M(\mathbf{k}, t)$ develops peaks as function of $\mathbf{k}$ whenever $\mathbf{q} \equiv \mathbf{k} - \mathbf{g}_{\mathbf{k}} = \mathbf{0}$. The values of $M(\mathbf{k}, t)$ at those peaks, once they have developed, are independent of time. This is because there is a zero eigenvalue $\lambda_{00} = 0$ with a nonzero weight $|\tilde{\psi}_{00}(\mathbf{g}_{\mathbf{k}})|^2 \neq 0$, which becomes the only important term in Eq. (7.63) for sufficiently long times. At other values of $\mathbf{k}$, all the eigenvalues with nonzero weights are greater than zero. Therefore, $M(\mathbf{k}, t)$ continues to decay with increasing t . (d) When t is large enough so that the above mentioned peaks develop, it is also large enough that the lowest eigenvalue $\lambda_{0\mathbf{q}}$ becomes the dominant term in Eq. (7.63) for any other $\mathbf{k}$. In that case, a minimum of $M(\mathbf{k}, t)$ usually appears whenever $\mathbf{q} \equiv \mathbf{k} - \mathbf{g}_{\mathbf{k}}$ approaches the edge of the first Brillouin zone, because $\lambda_{0\mathbf{q}}$ has a maximum there. The position of this minimum is close to but not exactly at the zone edge, because the weight $|\tilde{\psi}_{0\mathbf{q}}(\mathbf{g}_{\mathbf{k}})|^2$ also depends upon $\mathbf{q}$. (e) We can see from Fig. 7.5 that the lowest eigenvalue is the only one that needs

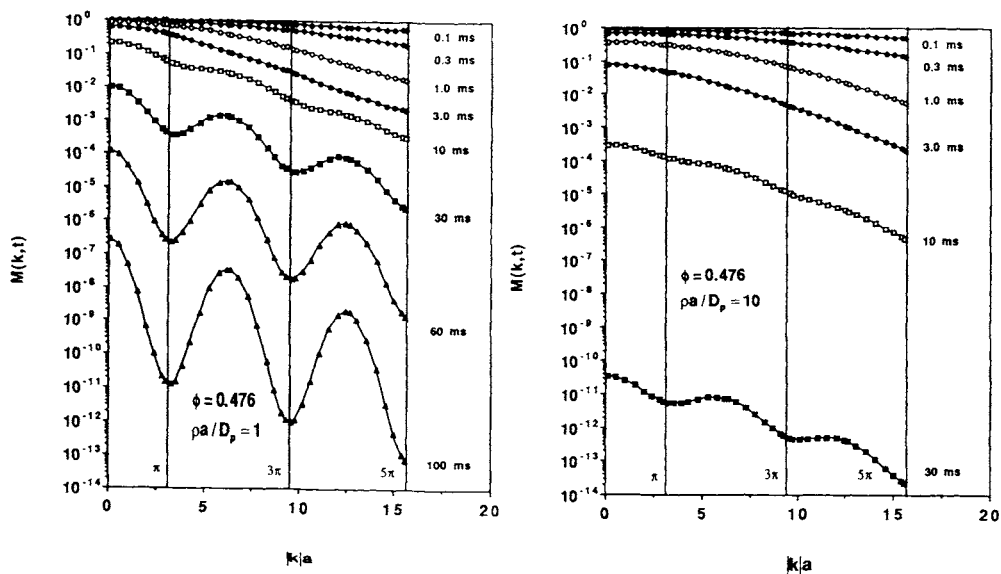


Figure 7.6: The PFGSE amplitude $M(\mathbf{k}, t)$ plotted vs. $|\mathbf{k}|a$ along the (100) direction at various times for a cubic array of touching spheres ($\phi = 0.476$) (a) for $\rho a / D_p = 1$ (b) for $\rho a / D_p = 10$ (after Fig. 6 of Ref. [18]).

to be considered as soon as $t \geq 0.01$ sec. That is $1/4$ of the natural time scale of this diffusion problem $a^2/D_p = 0.04$ sec, obtained using $a = 10 \mu\text{m}$ for the unit cell size and $D_p = 2.5 \times 10^{-5} \text{ cm}^2/\text{sec}$. For such times the values of $M(\mathbf{k}, t)$ at the minima quickly become much smaller than the detection capability of any practical measuring device.

Fig. 7.6 exhibits qualitatively similar behavior for systems with nonzero ρ , but the oscillations of $M(\mathbf{k}, t)$ are increasingly damped with increasing ρ . The overall decay of $M(\mathbf{k}, t)$ with increasing time also increases with ρ .

Another feature which sometimes appears can be seen in Figs. 7.7 and 7.8, which show results for the same systems as in Figs. 7.5 and 7.6(a), but now with $\mathbf{k} \parallel (110)$. Near the middle of the second Brillouin zone, a sharp minimum appears near the position where a maximum is trying to develop at $\mathbf{k} = \mathbf{g}_k$ with increasing t . The sharp minimum is caused by the fact that the weight factor $|\tilde{\psi}_{0\mathbf{q}}(\mathbf{g}_k)|^2$ of the lowest eigenvalue band vanishes for some small value of $\mathbf{q} = \mathbf{k} - \mathbf{g}_k$, i.e., for $\mathbf{k}$ very near to $\mathbf{g}_k$. This feature persists even when $\rho \neq 0$. In the case where $\rho = 0$, we have $\tilde{\psi}_{00}(\mathbf{g}) \propto \theta_{\mathbf{g}}$, therefore this feature is connected to the fact that $\theta_{\mathbf{g}_k}$ is very small. Clearly,

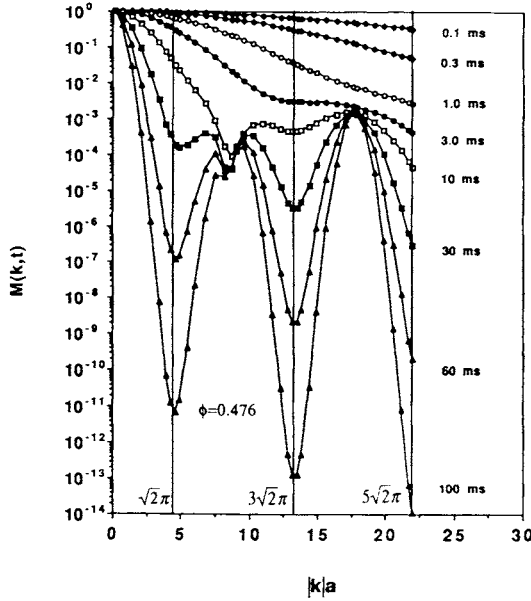


Figure 7.7: The PFGSE amplitude $M(\mathbf{k}, t)$ as function of $|\mathbf{k}|a$ along the (110) direction at various times for a simple cubic array of identical touching spheres with $\phi = 0.476$ and $\rho = 0$ (after Fig. 11 of Ref. [77]).

such an occurrence depends on the detailed shape of the pore space. This demonstrates the sensitivity of $M(\mathbf{k}, t)$ to those details, and indicates its potential usefulness for obtaining information about them.

7.2.2 Return to the origin probability (RTOP)

In Sections 7.1.2, 7.1.3 we saw that measurements of the time dependent diffusion coefficient $D(t)$ can provide information about the microstructure of the porous medium. That information came in the form of the value of S_p/V_p (from the short time behavior), the value of D_e (from the long time behavior), and the spectrum of relaxation rates (from the intermediate time behavior).

Another measurable quantity, with a similar potential for providing microstructural information about the porous medium, is the integrated return-to-the-origin probability (RTOP), defined as

$$P(t) \equiv \int_{V_p} dV G(\mathbf{r}, \mathbf{r}|t). \quad (7.68)$$

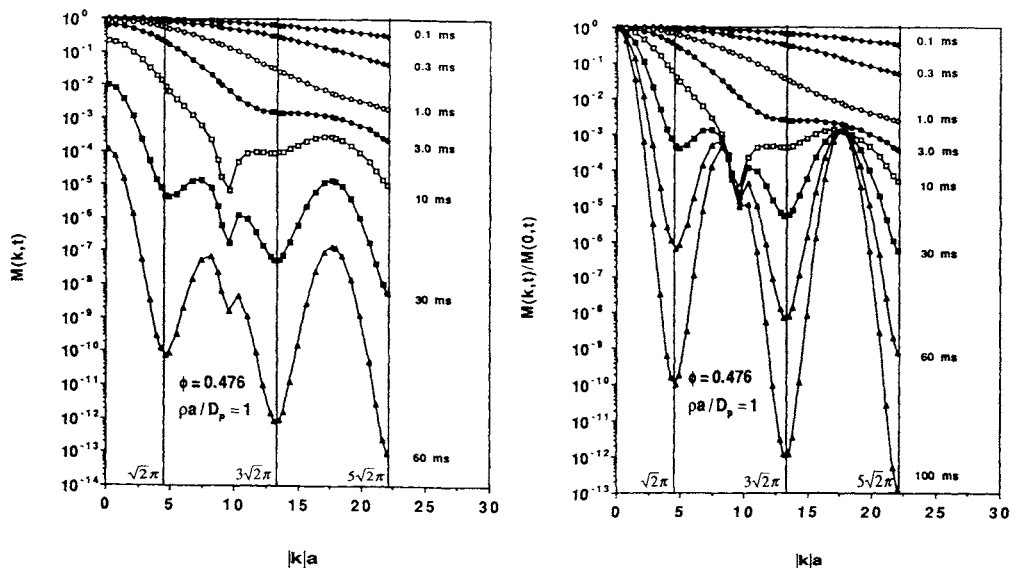


Figure 7.8: Plots of (a) $M(\mathbf{k}, t)$ and (b) $M(\mathbf{k}, t)/M(0, t)$ vs. $|\mathbf{k}|a$ along the (110) direction at various times for a cubic array of touching spheres ($\phi = 0.476$) and $\rho a/D_p = 1$ (after Fig. 8 of Ref. [18]).

This definition is based upon the fact that $G(\mathbf{r}, \mathbf{r}|t)dV$ is the conditional probability of a particle to be found in the infinitesimal volume element dV near the point $\mathbf{r}$, at the time t , if it was at that same point $\mathbf{r}$ at time $t = 0$. When this is integrated over the entire pore space V_p , it becomes the *average* probability of any particle returning to its starting position after a time t . We shall see that the RTOP depends on the microstructure in ways that enable us to extract information about the microgeometry of the porous medium.

A more general concept is the average probability density for any particle to advance by the vector displacement $\mathbf{R}$ after a time t . That is the following volume average of the diffusion propagator (this is the same as the ensemble average in the case of a disordered microstructure), and is given by

$$P(\mathbf{R}, t) \equiv \frac{1}{V_p} \int dV G(\mathbf{R} + \mathbf{r}, \mathbf{r}|t) = \int \frac{d^3k}{(2\pi)^3} M(\mathbf{k}, t) e^{i\mathbf{k} \cdot \mathbf{R}}.$$

It is a function of t and of the “relative coordinate” $\mathbf{R}$. $P(\mathbf{R}, t)$ always starts out with the simple, well defined, delta function form

$$P(\mathbf{R}, t = 0) = \delta^3(\mathbf{R}),$$

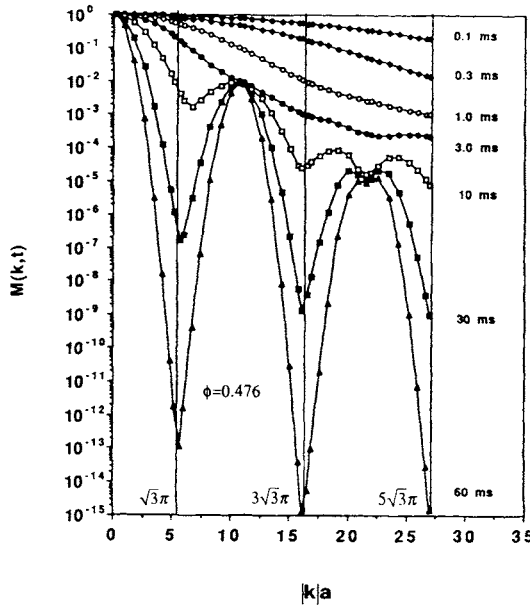


Figure 7.9: The PFGSE amplitude $M(\mathbf{k}, t)$ as function of $|\mathbf{k}|a$ along the (111) direction at various times for a simple cubic array of identical touching spheres with $\phi=0.476$ and $\rho = 0$ (after Fig. 12 of Ref. [77]).

and ends up as $P(\mathbf{R}, t \rightarrow \infty) = 0$ for a connected pore space, or when there are nonzero spin absorption coefficients.

When $\mathbf{R} = \mathbf{0}$, this is the same as the integrated return to the origin probability (RTOP) $P(t)$, divided by V_p . We conclude that $P(t = 0)$ is infinite. However, for all $t > 0$, $P(t)$ is a finite, well defined quantity, which can be obtained from $M(\mathbf{k}, t)$:

$$P(t) = V_p P(\mathbf{R} = \mathbf{0}, t) = V_p \int \frac{d^3 k}{(2\pi)^3} M(\mathbf{k}, t) = \sum_{\lambda} e^{-\lambda t}. \quad (7.69)$$

The last result follows from Eqs. (7.26) and (7.29), and shows that the RTOP is a “spectral function” of the diffusion problem in the porous medium: It is equal to a simple sum of all the characteristic exponential decay factors of the eigenstates of this problem.

If one could measure the RTOP with sufficient accuracy, one would be able to extract all the characteristic decay rates λ . It is amazing that such a simple arithmetic sum of exponentials also has the vivid physical interpretation of

being the integrated probability of finding a diffusing particle, after time t , back at its starting point.

As pointed out in Ref. [110], the short time behavior of this quantity contains information about the important microstructural parameter S_p/V_p —see Eq. (7.71) below. We shall also see below that, at long times, the RTOP includes information about D_e and ϕ [192]. As in the case of $D(t)$, one needs to know the RTOP at intermediate times in order to get more detailed information about the microstructure and the spectrum of decay rates.

In practice, the RTOP of a fluid filled porous medium can be determined by integrating $M(\mathbf{k}, t)$, either measured in a PFGSE experiment or calculated as described in Section 7.2.1 and Appendix A, over all $\mathbf{k}$, as in Eq. (7.69).

Renormalized RTOP

In a uniform fluid, since $G_0(\mathbf{r}, \mathbf{r}'|t)$ depends only on $|\mathbf{r} - \mathbf{r}'|$ [see Eq. (7.23)], $P(t)$ has the following simple form

$$P_0(t) = VG_0(\mathbf{r}, \mathbf{r}|t) = \frac{V}{(4\pi D_p t)^{3/2}},$$

where we ignore the uniform exponential decay rate $e^{-(1/T_2 + \nu)t}$. Note that we use the subscript 0 to denote quantities related to an infinite uniform fluid. Other quantities refer to a porous medium.

As shown earlier, $P(t = 0) = \infty$ —this behavior is also exhibited by the uniform fluid function $P_0(t)$. Also, when $t \rightarrow \infty$, $P(t)$ will tend to 0 if the pore space is connected or if at least one of the particle absorption coefficients ν , ρ is nonzero. In order to factor out $(4\pi D_p t)^{-3/2}$, as well as the intrinsic decay rate of $M(t)$ and its decay due to volume and surface sinks, it is useful to define a renormalized RTOP as follows:

$$P_s(t) \equiv \frac{P(t)M(0)}{P_0(t)M(t)}. \quad (7.70)$$

This quantity starts out as $P_s(0) = 1$, and ends up at a *higher finite value* as $t \rightarrow \infty$, even when both volume and surface absorption coefficients are nonzero, and even when the pore space is connected. In contrast with $D(t)$, which appears to be a monotonic decreasing function of t in all the numerical and asymptotic calculations, $P_s(t)$ has a *non-monotonic dependence* on t : It starts out by first increasing to values greater than 1

$$P_s(t) \cong 1 + \frac{1}{2} \frac{S_p}{V_p} \sqrt{\pi D_p t} \quad \text{for } pt \ll \sqrt{D_p t} \ll a. \quad (7.71)$$

At long times it tends to its asymptotic value, given by [192, 21]

$$P_s(\infty) = \frac{1}{\phi} \left(\frac{D_p}{D_e} \right)^{3/2} \frac{V_p \int_{V_p} dV \psi_0^2(\mathbf{r})}{\left[\int_{V_p} dV \psi_0(\mathbf{r}) \right]^2} > 1, \quad (7.72)$$

where $\psi_0(\mathbf{r})$ is again the eigenfunction of the diffusion problem in the pore space corresponding to the lowest eigenvalue or decay rate see Eqs. (7.33), (7.34). However, the approach to that asymptotic value is always *from above*. These statements have been proven rigorously, and are also verified by numerical computations [192]. The non-monotonic behavior of $P_s(t)$ appears for $\rho = 0$ as well as for $\rho \neq 0$, but it appears to be stronger in the latter case, and is especially strong when the porosity is low. This is evident in Fig. 7.10, which is taken from that reference. Fig. 7.11 shows the lower ϕ , $\rho = 0$ results of Fig. 7.10 once again, together with results of random walker simulations and short time asymptotics for comparison.

At intermediate times, Eq. (7.69) shows that $P(t)$ depends on the entire spectrum of relaxation rates. Therefore measurements of $P(t)$ or $P_s(t)$, as function of t for such times, can provide detailed information about that spectrum, and consequently about the detailed microstructure of the pore space.

Comparison with $D(t)$

The quantities $D(t)/D_p - 1$ and $P_s(t) - 1$ have much in common. At short times, when $t \rightarrow 0$, they are both proportional to $\sqrt{D_p t S_p/V_p}$:

$$\begin{aligned} \frac{D(t)}{D_p} - 1 &= -\frac{4}{9} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} + \mathcal{O}(D_p t), \\ P_s(t) - 1 &= \frac{\pi}{2} \frac{S_p}{V_p} \left(\frac{D_p t}{\pi} \right)^{1/2} + \mathcal{O}(D_p t). \end{aligned}$$

The opposite signs in the leading term of the two expressions can be easily understood: The diffusion of particles in the pore space is hindered by the solid matrix, which *reduces* their macroscopic rate of displacement away from their starting position, while *increasing* the return to the origin probability (RTOP).

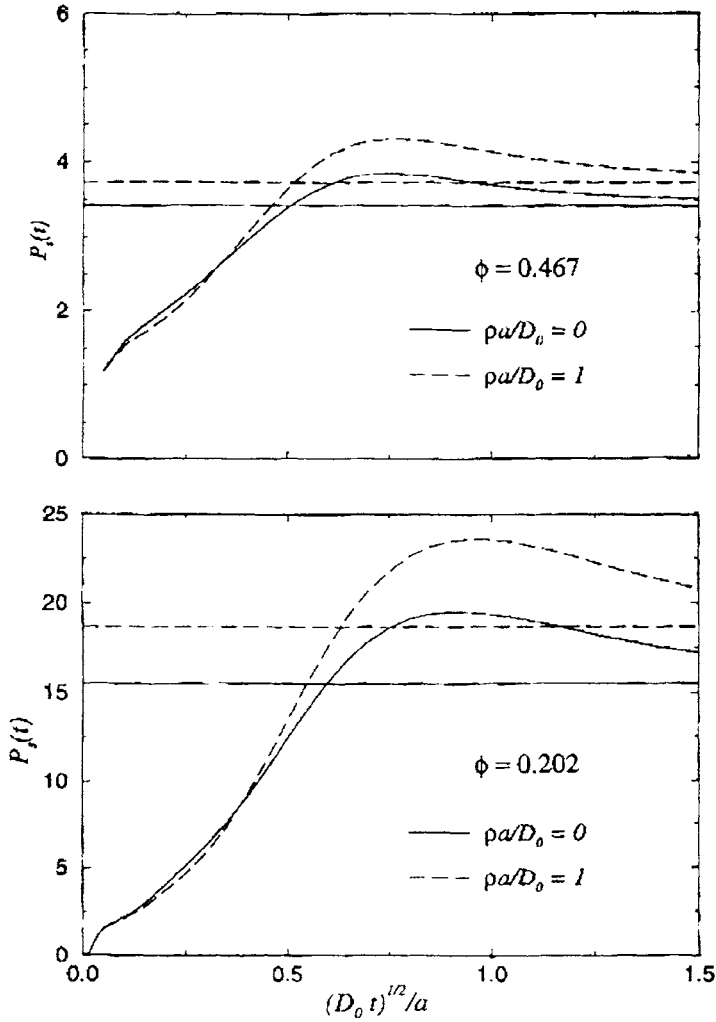


Figure 7.10: The effects of surface relaxation on the RTOP are illustrated for the touching sphere (upper panel) and overlapping sphere (lower panel) arrays of simple cubic packings. Note that, to leading order, the short time behavior of $P_s(t)$ is *not* affected by ρ . The horizontal straight lines represent the long time asymptotic result of Eq. (7.72) (after Fig. 8 of Ref. [192]).

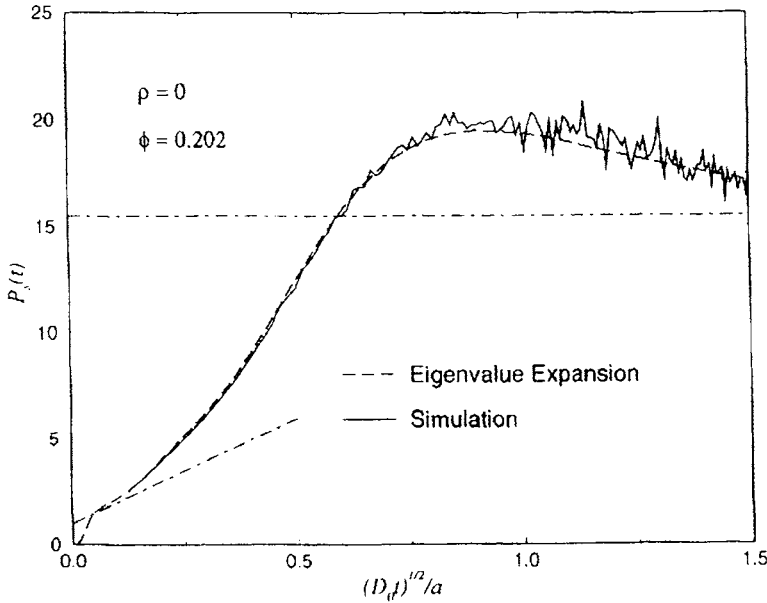


Figure 7.11: For a consolidated simple cubic packing of solid spheres (i.e., neighboring solid spheres overlap partially) with $\rho = 0$, eigenvalue expansion results for $P_s(t)$ are compared to partially smoothed data based on singular-value-decomposition analysis of the corresponding random walk simulations. Note that the random walk results fluctuate even after that drastic smoothing procedure. The lower (slanted) dot-dash line represents the short time asymptotic result of Eq. (7.71), while the upper (horizontal) dot-dash line represents the long time asymptotic result of Eq. (7.72) (after Fig. 4 of Ref. [192]).

At long times, both $D(t)$ and $P_s(t)$ tend to constant asymptotic values, which depend only on the lowest lying eigenstates [see Eqs. (7.35) and (7.72)]:

$$D(t) \rightarrow D_e = D_p \frac{\int_{V_p} dV \psi_0^2 (\nabla \phi_\alpha)^2}{\int_{V_p} dV \psi_0^2}, \quad P_s(t) \rightarrow \frac{1}{\phi} \left(\frac{D_p}{D_e} \right)^{3/2} \frac{V_p \int_{V_p} dV \psi_0^2(\mathbf{r})}{\left[\int_{V_p} dV \psi_0(\mathbf{r}) \right]^2}.$$

All of the preceding considerations lead to the conclusion that the asymptotic behavior of a diffusing particle in a porous medium, at times that are either very short or very long, can provide only limited information about the microstructure. In order to obtain detailed information, the behavior of both $D(t)$ and $P_s(t)$ at intermediate times (which depend on the entire

spectrum of eigenstates) must be measured, and their dependence on the microstructure topology must be studied and understood.

7.3 Random walker simulations

Diffusive motion of a particle is the result of many interactions which take place, between that particle and other neighboring particles, in an *incoherent fashion*, i.e., with rapid loss of memory as time progresses. This is often described by the simple model of “uncorrelated discrete random walk”, in which time advances by a finite step τ (τ can either be fixed or variable from step to step), and the particle position at time $t + \tau$ is chosen at random from a set of possible positions according to a probability distribution.

Various implementations of this model have been used in computer simulations of a physical diffusion process. Such simulations mimic the actual microscopic processes in a simplified manner. Nevertheless, they allow us to follow the microscopic motion of the particle with some finite small scale of resolution.

Random walker simulations of diffusive motion have the advantage that, in contrast with solutions of the differential equation for the spin density, they are able to deal with a complex and disordered pore space almost as easily as when that space has a simple ordered shape.

The main disadvantage is that the simulation can proceed only for times that are not excessively long compared to the basic time step τ . In simulations where D_e is much less than D_p , either as a result of bottlenecks in the pore space or as a result of strong absorption at the pore/matrix interface, the limitation to short times has particularly disastrous effects on the attainable results: When narrow necks or cracks play a dominant role, it takes a very long time for the random walker to explore those elements of the pore space, and for their full impact to affect macroscopic quantities like $M(\mathbf{k}, t)$, $D(t)$, and $P(t)$. When ρ is large, many walkers get wiped out before they can explore the detailed shape of the pore space. Thus, many more walkers must be simulated in order to determine those macroscopic quantities.

Nevertheless, random walker simulations can be carried out to times that are well beyond those where short time asymptotics are valid—see Appendix B. They therefore allow quantitative comparisons to be made, not only with those asymptotics, but also with results based on numerical calculations of the diffusion eigenstates, which are valid only for times that are not too

short—see Appendix A. The latter correspondence is especially important: On one hand, it allows the parameters of the random walk simulations to be chosen so as to correctly reproduce the results of the diffusion equation. On the other hand, by comparing random walk simulations on periodic and non-periodic pore structures, which have common elements like the typical pore sizes, narrow necks, and narrow cracks, we can determine the extent to which periodic models of the pore structure are able to mimic the macroscopic behavior of the disordered pore space found in real rocks.

Infinite fluid

The simplest computational implementation of a three dimensional random walk in an unrestricted geometry has a fixed time step τ , and a fixed simple cubic lattice of discrete possible positions in space, with lattice constant ϵ . Starting at some given position, the walker advances to one of the nearest neighbor positions, chosen at random and with equal probabilities, as the time advances by τ . After N time steps, the walker has reached a position some distance away from the origin. A simple statistical analysis shows that the *root mean square distance* R is given, for large N , by $R = \sqrt{N}\epsilon$. Since the total time elapsed is $t = N\tau$, we can determine D_p by the following expression, which is based upon Eq. (7.24):

$$D_p = \frac{R^2}{6t} = \frac{\epsilon^2}{6\tau}. \quad (7.73)$$

Note that the number of discrete time steps N has dropped out of the final result. It is worth noting that, in this simple model, all the above results are valid for *arbitrary values of N* , i.e., even if the number of steps N is as small as 1.

Restricted geometry

In a restricted geometry or pore space, the walker sometimes encounters a boundary, where it is either reflected or wiped out (mimicking interface absorption of the particle or depolarization of the spin). This can be implemented by modifying the above described algorithm as follows: When the walker is at a boundary site and tries to move into the boundary (i.e., in a direction where the nearest neighbor site is absent), it is either wiped out with probability γ , or it stays put at the boundary site with probability $1 - \gamma$, even though the time advances by τ .

The various quantities $M(t)/M(0)$, $M(\mathbf{k}, t)$, $D(t)$ can now be calculated from the discrete paths $\mathbf{r}_n(t)$, $n = 1 \dots N$ followed by a sufficiently large number N of simulated random walkers: $M(t)/M(0)$ is simply equal to the fraction of walkers that survive at time t . The PFGSE amplitude is found from

$$M(\mathbf{k}, t) = \frac{1}{N} \sum_{n=1}^N e^{i\mathbf{k} \cdot [\mathbf{r}_n(t) - \mathbf{r}_n(0)]}.$$

The time dependent diffusion coefficient can be found from this, using Eq. (7.37) or Eq. (A.16). However, it is easier to use the more physical definition of that coefficient, expressed by Eq. (A.14), to get

$$6D(t)t = \frac{1}{N} \sum_{n=1}^N [\mathbf{r}_n(t) - \mathbf{r}_n(0)]^2. \quad (7.74)$$

Note, however, that those two definitions of $D(t)$ lead to different results when $\rho \neq 0$ —see the discussion which follows Eq. (A.15). Finally, $P(t)$ is found by integrating $M(\mathbf{k}, t)$ over all $\mathbf{k}$, as described in Eq. (7.69) [192].

The relation between the continuum diffusion coefficient of the pore fluid D_p and the random walk parameters ϵ , τ , i.e., Eq. (7.73), has a very wide range of validity: It extends to other implementations of a random walk, e.g., to implementations where the walker position after time τ is not limited to a discrete set of nearest neighbors but can lie anywhere on a spherical surface of radius ϵ around the origin. It also extends to an implementation where the spatial step size ϵ and the time step τ are changed from step to step: At each step, ϵ is made as large as possible without getting any intersections of that spherical surface with any boundary or interface. At the same time, τ is also changed so as to keep satisfying Eq. (7.73) with a fixed value of D_p . It turns out that, as long as these restrictions are maintained, the random walk mimics the diffusion process faithfully [193]. The latter implementation can greatly accelerate the computation by allowing large steps to be used, whenever possible, in the random walker simulation.

Mimicking interface absorption

In contrast with the simplicity and widespread validity of Eq. (7.73), the relation between the continuum interface absorption coefficient ρ and the probability γ , that the walker is wiped out when it reaches the boundary of

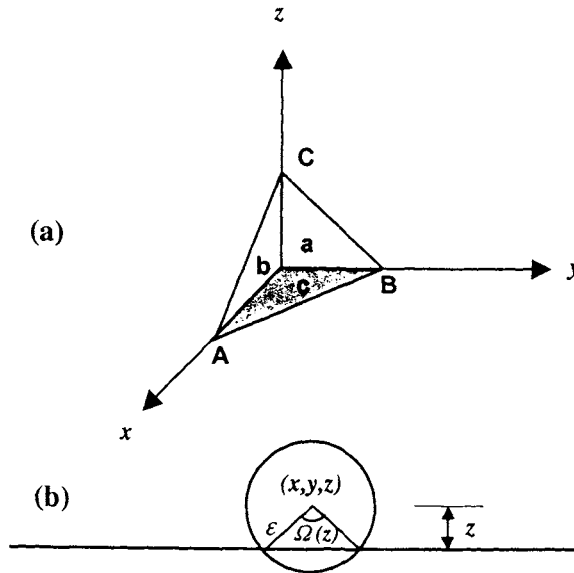


Figure 7.12: (a) Illustration of the surface area enhancement which results from a discretization of space in a random walker simulation. (b) Illustration of the configuration used to calculate the probability for the random walker to be wiped out using a random walk implementation where the walker jumps to an arbitrary point on the surface of a sphere of radius ϵ , as described briefly in the text, and in greater detail in Section IIA of Ref. [20] (after Fig. 6 of Ref. [20]).

the pore space, involves more subtle reasoning. This is most readily appreciated when the sites attainable by that walker are restricted to lie on a fixed simple cubic lattice. This spatial discretization has the effect of changing most smooth interfaces into a collection of right-angled corners. This is evident if we consider a situation where the interface intersects three neighboring orthogonal edges or bonds of the lattice—see Fig. 7.12(a). The discretization thus has the effect of increasing the area of the smooth interface by a factor that *depends on the local orientation of the interface* and is usually *significantly greater than 1*.

Other implementations of the random walk also enhance the interface area, by an amount whose average is usually non-negligible and depends on details of the implementation—see, e.g., Fig. 7.12(b). The consequent enhancement factors of the total interface area *do not tend to 1 as the step*

size of discretization is decreased, but to a value significantly greater than 1.

For an interface which is not only macroscopically isotropic, but is made up entirely of closed spherical surfaces, the value of this enhancement factor in the discrete space implementation, averaged over all possible orientations of the interface, is $3/2$. Thus the relation between γ and ρ is found to be [20]

$$\gamma = \frac{2}{3} \frac{\epsilon \rho}{D_p}.$$

When these subtleties are taken into consideration, random walker simulations provide an important tool for studying diffusion in a porous medium. In particular, they are able to study disordered pore structures and, for periodic microstructures, they are able to bridge any gap that remains between short time asymptotics and eigenstate based calculations, whose validity breaks down at short times. Figs. 7.13 and 7.14 show results for $M(\mathbf{k}, t)$, calculated for periodic simple cubic arrays of spherical obstacles of radius a using both methods, with different values of ϕ and with $\rho a/D_p = 0, 0.2$, and 1. A similar comparison for $D(t)$ appears in Fig. 7.15, and a similar comparison for the normalized return-to-the-origin-probability-density (RTOP) $P_s(t)$ appears in Fig. 7.11. These comparisons lend confidence in both computational methods, whose ranges of validity are largely complementary.

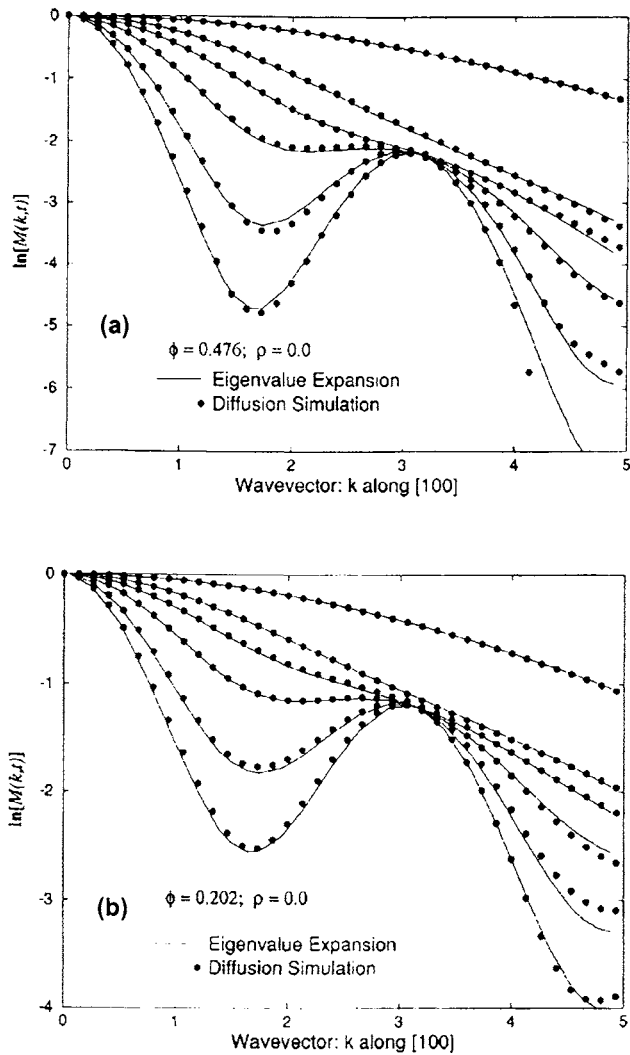


Figure 7.13: Eigenstate expansion and random walker simulation results for $M(\mathbf{k}, t)$, in a cubic array of spheres microstructure, are compared in the case $\rho = 0$ (a) for touching spheres ($\phi = 0.4674$) (b) for overlapping spheres ($\phi = 0.202$). Results are presented for six values of t corresponding to 0.67, 3.33, 6.67, 13.33, 26.67 and 40 msec, in descending order of the curves. The cubic unit cell edge is taken to be $10 \mu\text{m}$ and $D_p = 2.5 \mu\text{m}^2/\text{msec}$ (after Fig. 2 of Ref. [20]).

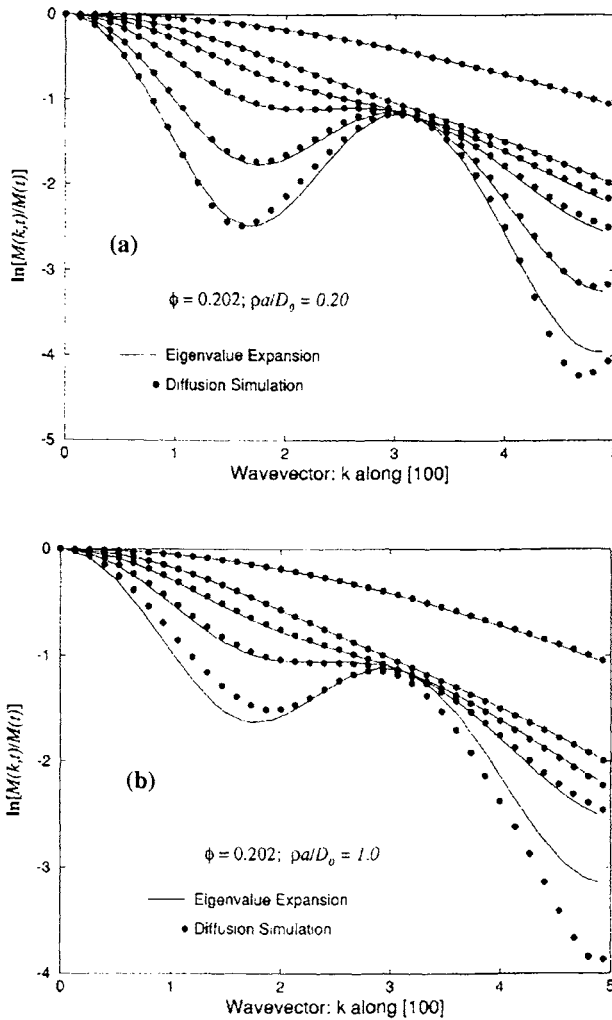


Figure 7.14: Comparison of results for $M(\mathbf{k}, t)/M(t)$, using the eigenstate expansion and random walker simulations, in the same microstructure as that of Fig. 7.13. In this comparison, two different nonzero values were assumed for the relaxation strength ρ : In (a) $\rho a/D_p = 0.20$ and $t = 0.67 \rightarrow 40$ msec, while in (b) $\rho a/D_p = 1.0$ and $t = 0.67 \rightarrow 26.67$ msec. In both sets of calculations, $\phi = 0.202$; in the figure D_p is denoted by D_0 (after Fig. 4 of Ref. [20]).

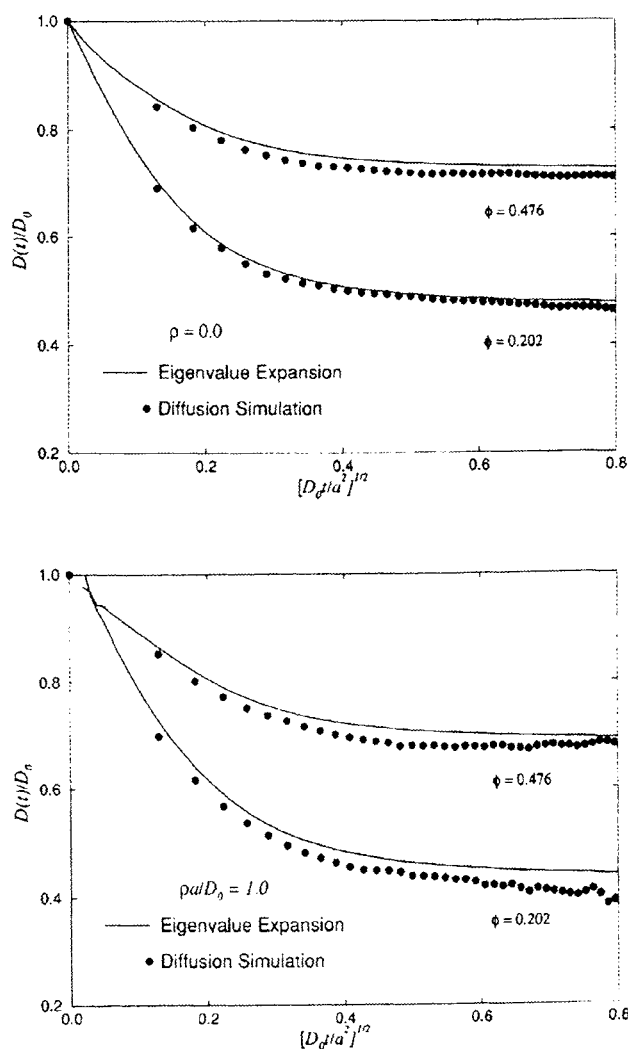


Figure 7.15: Results for $D(t)/D_p$ in the cubic array of spheres microstructure, based on the eigenstate expansion and on random walker simulations, are compared for two values of the relaxation strength ρ at the two porosities, $\phi = 0.4764$ and 0.202 . In (a) $\rho = 0$ while in (b) $\rho a/D_0 = 1.0$; in the figure D_p is denoted by D_0 (after Fig. 5 of Ref. [20]).